

严镇军《复变函数》习题全解

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1 复变数函数

1. 记 $z = x + iy$, 则 $w = \frac{1}{z} = \frac{x - iy}{x^2 + y^2}$. 再设 $w = u + iv$, 则 $u = \frac{x}{x^2 + y^2}, v = \frac{-y}{x^2 + y^2}$.

(1) $x = 1 \implies u = \frac{1}{1 + y^2}, v = \frac{-y}{1 + y^2} \implies u^2 + v^2 = \frac{1}{1 + y^2} = u$. 此时该曲线为以 $z_C = \frac{1}{2}$ 为圆心、 $R = \frac{1}{2}$ 为半径的圆周.

(2) $y = 0 \implies u = \frac{1}{x}, v = 0$. 此时该曲线为实轴.

(3) $y = x \implies u = \frac{1}{2x}, v = \frac{-1}{2x} \implies u + v = 0$. 此时该曲线为直线 $u + v = 0$.

(4) $x^2 + y^2 = 4 \implies u = \frac{x}{4}, v = \frac{-y}{4} \implies u^2 + v^2 = \frac{x^2 + y^2}{16} = \frac{1}{4}$. 此时该曲线为以原点为圆心、 $R = \frac{1}{2}$ 为半径的圆周.

(5) 这里 x, y 不好直接消元, 可逆向变换: $z = \frac{1}{w} \implies x = \frac{u}{u^2 + v^2}, y = \frac{-v}{u^2 + v^2}$, 则

$$5 = (x - 1)^2 + y^2 = \left(\frac{u}{u^2 + v^2} - 1 \right)^2 + \left(\frac{-v}{u^2 + v^2} \right)^2 = \frac{1}{u^2 + v^2} - \frac{2u}{u^2 + v^2} + 1,$$

即

$$u^2 + v^2 = \frac{1 - 2u}{4} \implies \left(u + \frac{1}{4} \right)^2 + v^2 = \frac{5}{16}.$$

此时该曲线为以 $z_C = -\frac{1}{4}$ 为圆心、 $R = \frac{\sqrt{5}}{4}$ 为半径的圆周.

2. 取路径 $y = kx$, 则 $z \neq 0$ 时, $f(z) = \frac{k}{1 + k^2}$, 因此 $\lim_{z \rightarrow 0} f(z)$ 与 k 的取值相关, 极限不存在, 故 $f(z)$ 在 $z = 0$ 处不连续.

3. 记 $p_n(z) = \sum_{k=0}^n a_k z^k$ ($a_n \neq 0$), 则

$$\begin{aligned} |p_n(z)| &= \left| \sum_{k=0}^n a_k z^k \right| \geq |a_n z^n| - \left| \sum_{k=0}^{n-1} a_k z^k \right| \geq |a_n z^n| - \sum_{k=0}^{n-1} |a_k| |z^k| \\ &= |z|^n \left(|a_n| - \sum_{k=0}^{n-1} \frac{|a_k|}{|z|^{n-k}} \right). \end{aligned}$$

因此当 $z \rightarrow \infty$ 即 $|z| \rightarrow +\infty$ 时, 不等号右侧为 $+\infty \cdot |a_n|$, 即 $\lim_{z \rightarrow \infty} |p_n(z)| = +\infty$, 即 $\lim_{z \rightarrow \infty} p_n(z) = \infty$.

4. (1) $f(z) = |z| = \sqrt{x^2 + y^2}$, 即 $u = \sqrt{x^2 + y^2}, v = 0$, 显然对 $\forall z \in \mathbb{C}$ 不满足 C-R 方程, $f(z)$ 在全平面处处不可导.

(2) $f(z) = x + y$, 即 $u = x + y, v = 0$, 显然对 $\forall z \in \mathbb{C}$ 不满足 C-R 方程, $f(z)$ 在全平面处处不可导.

$$(3) f(z) = \frac{1}{\bar{z}} = \frac{x+iy}{x^2+y^2}, \text{ 即 } u = \frac{x}{x^2+y^2}, v = \frac{y}{x^2+y^2}, \text{ 注意到}$$

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{y^2 - x^2}{(x^2 + y^2)^2}, & \frac{\partial v}{\partial y} &= \frac{x^2 - y^2}{(x^2 + y^2)^2}, \\ \frac{\partial u}{\partial y} &= -\frac{2xy}{(x^2 + y^2)^2}, & \frac{\partial v}{\partial x} &= -\frac{2xy}{(x^2 + y^2)^2}, \end{aligned}$$

即满足 C-R 方程当且仅当 $x^2 = y^2, xy = 0$ 同时成立 $\implies x = y = 0$. 但 $z = 0$ 时 $\bar{z} = 0$, $f(z)$ 无定义且在该处不连续, 故 $f(z)$ 在全平面处处不可导.

5. (1) $u = xy, v = y$, 由 C-R 方程得, $f(z)$ 可导时 $y = 1, x = 0$. 但 $f(z)$ 在 $z_0 = i$ 的邻域内不可导, 因此 $f(z)$ 在全平面都不解析.

(2) i. $|z| < 1$ 时, $f(z) = |z|z \implies u = x\sqrt{x^2+y^2}, v = y\sqrt{x^2+y^2}$, 由 C-R 方程, $f(z)$ 可导时

$$\frac{2x^2+y^2}{\sqrt{x^2+y^2}} = \frac{x^2+2y^2}{\sqrt{x^2+y^2}}, \quad \frac{xy}{\sqrt{x^2+y^2}} = -\frac{xy}{\sqrt{x^2+y^2}}.$$

从中解得 $x = y = 0$, 但 $f(z)$ 在原点附近不可导, 因此 $f(z)$ 在 $|z| < 1$ 处不解析.

ii. $|z| > 1$ 时, $f(z) = z^2$ 为幂函数, 显然解析.

iii. $|z| = 1$ 时, 由于 $f(z)$ 在 $|z| = 1$ 上的点的邻域中 $|z| < 1$ 的部分不解析, 因此 $f(z)$ 在 $|z| = 1$ 处必然不解析.

综上, $f(z)$ 的解析区域为 $|z| > 1$.

6. (1) $z^3 = (x+iy)^3 = (x^3 - 3xy^2) + i(3x^2y - y^3) \implies u = x^3 - 3xy^2, v = 3x^2y - y^3$, 则

$$\frac{\partial u}{\partial x} = 3(x^2 - y^2) = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -6xy = -\frac{\partial v}{\partial x}$$

对 $\forall x, y \in \mathbb{R}$ 成立, 因此 z^3 在全平面上解析.

(2) $u = e^x(x \cos y - y \sin y), v = e^x(y \cos y + x \sin y)$, 则

$$\frac{\partial u}{\partial x} = e^x[(x+1) \cos y - y \sin y] = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -e^x[(x+1) \sin y + y \cos y] = -\frac{\partial v}{\partial x}$$

对 $\forall x, y \in \mathbb{R}$ 成立, 因此该函数在全平面上解析.

(3) $u = \cos x \cosh y, v = -\sin x \sinh y$, 则

$$\frac{\partial u}{\partial x} = -\sin x \cosh y = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = \cos x \sinh y = -\frac{\partial v}{\partial x}$$

对 $\forall x, y \in \mathbb{R}$ 成立, 因此该函数在全平面上解析.

7. 注意到 $z \neq z_0$ 时,

$$\frac{f(z)}{g(z)} = \frac{f(z) - f(z_0)}{g(z) - g(z_0)} = \frac{f(z) - f(z_0)}{z - z_0} \bigg/ \frac{g(z) - g(z_0)}{z - z_0},$$

等式两端同时取 $z \rightarrow z_0$, 由于 $f(z), g(z)$ 在 z_0 解析, 故

$$\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}$$

8. (1) 由 C-R 方程, $f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y} = 0$, 则

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} = 0.$$

因此 $u(x, y), v(x, y) = \text{const}$, 亦即 $f(z) = u + iv = \text{const}$.

(2) 由于 $f(z) = u + iv, \overline{f(z)} = u - iv$ 均解析, 由 C-R 方程,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = \frac{\partial(-v)}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = -\frac{\partial(-v)}{\partial x},$$

从中解得 u, v 各偏导数均为 0, 同样有 $u, v = \text{const} \implies f(z) = \text{const}$.

(3) $\text{Re} z = u = \text{const}$ 时, 由 C-R 方程知 v 的各偏导数均为 0, 则 $v = \text{const}$, 进而 $f(z) = \text{const}$.

(4) 同上一条.

(5) $|f(z)|^2 = u^2 + v^2 = \text{const}$, 两端分别对 x, y 求偏导, 约分得

$$\begin{cases} u \frac{\partial u}{\partial x} + v \frac{\partial v}{\partial x} = 0, \\ u \frac{\partial u}{\partial y} + v \frac{\partial v}{\partial y} = 0, \end{cases}$$

联立 C-R 方程, 消去 v 的各偏导得

$$\begin{cases} u \frac{\partial u}{\partial x} - v \frac{\partial u}{\partial y} = 0, \\ u \frac{\partial u}{\partial y} + v \frac{\partial u}{\partial x} = 0. \end{cases}$$

当 $u = v \equiv 0$ 时, 显然 $f(z) = 0$ 为常数; 否则从中解得 $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = 0 \implies u = \text{const}$, 同理 $v = \text{const}$, 进而 $f(z) = \text{const}$.

(6) 记 $\theta = \arg f(z) = \text{const}$, 则 $v = u \tan \theta = ku$, 其中 $k = \tan \theta = \text{const}$, 将其代入 C-R 方程:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = k \frac{\partial u}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = -k \frac{\partial u}{\partial x},$$

从中解得 $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = 0$; 特别地, 若 $\theta = \frac{\pi}{2}$ 即 $k = +\infty$, 则 $u = \frac{v}{k} = 0$, 由 C-R 方程知 $u = 0$. 因此无论 k 取何值, 均有 $f(z) = \text{const}$.

9. 由链式法则,

$$\begin{aligned} \frac{\partial H}{\partial x} &= \frac{\partial H}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial H}{\partial \eta} \frac{\partial \eta}{\partial x}, \\ \frac{\partial^2 H}{\partial x^2} &= \left(\frac{\partial^2 H}{\partial \xi^2} \frac{\partial \xi}{\partial x} + \frac{\partial^2 H}{\partial \eta \partial \xi} \frac{\partial \eta}{\partial x} \right) \frac{\partial \xi}{\partial x} + \frac{\partial H}{\partial \xi} \frac{\partial^2 \xi}{\partial x^2} + \left(\frac{\partial^2 H}{\partial \xi \partial \eta} \frac{\partial \xi}{\partial x} + \frac{\partial^2 H}{\partial \eta^2} \frac{\partial \eta}{\partial x} \right) \frac{\partial \eta}{\partial x} + \frac{\partial H}{\partial \eta} \frac{\partial^2 \eta}{\partial x^2} \\ &= \left[\frac{\partial^2 H}{\partial \xi^2} \left(\frac{\partial \xi}{\partial x} \right)^2 + \frac{\partial^2 H}{\partial \eta^2} \left(\frac{\partial \eta}{\partial x} \right)^2 \right] + 2 \frac{\partial^2 H}{\partial \xi \partial \eta} \frac{\partial \xi}{\partial x} \frac{\partial \eta}{\partial x} + \left(\frac{\partial H}{\partial \xi} \frac{\partial^2 \xi}{\partial x^2} + \frac{\partial H}{\partial \eta} \frac{\partial^2 \eta}{\partial x^2} \right), \end{aligned}$$

而 $\frac{\partial^2 H}{\partial y^2}$ 同理. 由 C-R 方程,

$$\frac{\partial \xi}{\partial x} \frac{\partial \eta}{\partial x} + \frac{\partial \xi}{\partial y} \frac{\partial \eta}{\partial y} = -\frac{\partial \xi}{\partial x} \frac{\partial \xi}{\partial y} + \frac{\partial \xi}{\partial y} \frac{\partial \xi}{\partial x} = 0.$$

注意到

$$f'(z) = \frac{\partial \xi}{\partial x} - i \frac{\partial \xi}{\partial y} = \frac{\partial \eta}{\partial y} + i \frac{\partial \eta}{\partial x} \implies |f'(z)|^2 = \left(\frac{\partial \xi}{\partial x}\right)^2 + \left(\frac{\partial \xi}{\partial y}\right)^2 = \left(\frac{\partial \eta}{\partial x}\right)^2 + \left(\frac{\partial \eta}{\partial y}\right)^2,$$

并且 $f(z)$ 有任意阶导数, 对于 $f'(z) = \frac{\partial \xi}{\partial x} + i \frac{\partial \eta}{\partial x} = \frac{\partial \eta}{\partial y} - i \frac{\partial \xi}{\partial y}$, 由 C-R 方程,

$$\frac{\partial^2 \xi}{\partial x^2} = \frac{\partial^2 \eta}{\partial x \partial y} = -\frac{\partial^2 \xi}{\partial y^2}, \quad \frac{\partial^2 \eta}{\partial x^2} = -\frac{\partial^2 \xi}{\partial x \partial y} = -\frac{\partial^2 \eta}{\partial y^2},$$

因此

$$\begin{aligned} \frac{\partial^2 H}{\partial x^2} + \frac{\partial^2 H}{\partial y^2} &= \left[\left(\frac{\partial \xi}{\partial x}\right)^2 + \left(\frac{\partial \xi}{\partial y}\right)^2 \right] \frac{\partial^2 H}{\partial \xi^2} + \left[\left(\frac{\partial \eta}{\partial x}\right)^2 + \left(\frac{\partial \eta}{\partial y}\right)^2 \right] \frac{\partial^2 H}{\partial \eta^2} \\ &\quad + 2 \left(\frac{\partial \xi}{\partial x} \frac{\partial \eta}{\partial x} + \frac{\partial \xi}{\partial y} \frac{\partial \eta}{\partial y} \right) \frac{\partial^2 H}{\partial \xi \partial \eta} \\ &\quad + \left(\frac{\partial^2 \xi}{\partial x^2} + \frac{\partial^2 \xi}{\partial y^2} \right) \frac{\partial H}{\partial \xi} + \left(\frac{\partial^2 \eta}{\partial x^2} + \frac{\partial^2 \eta}{\partial y^2} \right) \frac{\partial H}{\partial \eta} \\ &= |f'(z)|^2 \frac{\partial^2 H}{\partial \xi^2} + |f'(z)|^2 \frac{\partial^2 H}{\partial \eta^2} + 2 \cdot 0 \cdot \frac{\partial^2 H}{\partial \xi \partial \eta} + 0 \cdot \frac{\partial H}{\partial \xi} + 0 \cdot \frac{\partial H}{\partial \eta} \\ &= |f'(z)|^2 \left(\frac{\partial^2 H}{\partial x^2} + \frac{\partial^2 H}{\partial y^2} \right). \end{aligned}$$

10. 极坐标下, $x = r \cos \theta, y = r \sin \theta$, 因此

$$\begin{aligned} \frac{\partial u}{\partial r} &= \cos \theta \frac{\partial u}{\partial x} + \sin \theta \frac{\partial u}{\partial y}, \\ \frac{\partial v}{\partial r} &= \cos \theta \frac{\partial v}{\partial x} + \sin \theta \frac{\partial v}{\partial y}, \\ \frac{\partial u}{\partial \theta} &= r \left(-\sin \theta \frac{\partial u}{\partial x} + \cos \theta \frac{\partial u}{\partial y} \right), \\ \frac{\partial v}{\partial \theta} &= r \left(-\sin \theta \frac{\partial v}{\partial x} + \cos \theta \frac{\partial v}{\partial y} \right). \end{aligned}$$

利用 C-R 方程:

$$\begin{aligned} \frac{\partial u}{\partial r} &= \cos \theta \frac{\partial v}{\partial y} - \sin \theta \frac{\partial v}{\partial x} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \\ \frac{1}{r} \frac{\partial u}{\partial \theta} &= -\left(\sin \theta \frac{\partial v}{\partial y} + \cos \theta \frac{\partial v}{\partial x} \right) = -\frac{\partial v}{\partial r}. \end{aligned}$$

11. (1) $\frac{1}{w}, w = z^2 - 3z + 2$ 在有定义点均解析, 因此只需要除去无定义点: $z_1 = 1, z_2 = 2$, 其微商:

$$\frac{d}{dz} \left(\frac{1}{z^2 - 3z + 2} \right) = \frac{3 - 2z}{(z^2 - 3z + 2)^2}.$$

(2) 同上, 显然该函数在有定义点解析, 无定义点: $z^3 + a = 0 \implies z = \sqrt[3]{a} e^{i \frac{\arg a + 2k\pi}{3}}, k = 0, 1, 2$, 即

$$z_1 = \sqrt[3]{a}, \quad z_2 = \sqrt[3]{a} e^{i \frac{2\pi}{3}} = \frac{\sqrt[3]{a}}{2} (-1 + \sqrt{3}i), \quad z_3 = \sqrt[3]{a} e^{i \frac{4\pi}{3}} = -\frac{\sqrt[3]{a}}{2} (1 + \sqrt{3}i),$$

其微商:

$$\frac{d}{dz} \left(\frac{1}{z^3 + a} \right) = -\frac{3z^2}{(z^3 + a)^2}.$$

12. 显然多项式函数 w 在 $|z| < 1$ 内解析, 下证该映照为一一映照: 对 $\forall z_1, z_2 \in \mathbb{C}$, 其对应数值分别为 w_1, w_2 , 则

$$w_1 - w_2 = (z_1^2 - z_2^2) + 2(z_1 - z_2) = (z_1 - z_2)(z_1 + z_2 + 2).$$

当 $w_1 = w_2$ 时, 要求 $z_1 = z_2$ 或 $z_1 + z_2 = -2$.

若为第一种情况, 则已得证; 若为第二种情况, 注意到 $|z| < 1$, 即 $|z_1 + z_2| \leq |z_1| + |z_2| < 2$, 即第二个等式不可能成立. 综上, $w_1 = w_2$ 当且仅当 $z_1 = z_2$, 因此该函数为从 z 到 w 的单叶映照.

13. $w = \sqrt{z}$ 的两个单值连续分支: $w_k = \sqrt{|z|} e^{i \frac{\arg z + 2k\pi}{2}} (k = 0, 1)$, 具体为:

$$w_0 = \sqrt{|z|} e^{i \frac{\arg z}{2}}, \quad w_1 = \sqrt{|z|} e^{i (\frac{\arg z}{2} + \pi)} \left(-\frac{3\pi}{2} \leq \arg z < \frac{\pi}{2} \right).$$

由题意, $w(z) \big|_{\arg z = 2k'\pi (k' \in \mathbb{Z})} = 0$, 此时 $w_0 = \sqrt{|z|} > 0, w_1 = -\sqrt{|z|} < 0$, 因此取 w_0 . 故

$$w_0(i) \bigg|_{\arg z = -\frac{3\pi}{2}^+} = e^{-i \frac{3\pi}{4}} = -\frac{1+i}{\sqrt{2}}, \quad w_0(i) \bigg|_{\arg z = \frac{\pi}{2}^-} = e^{i \frac{\pi}{4}} = \frac{1+i}{\sqrt{2}}.$$

进而

$$w(-1) = e^{-i \frac{\pi}{2}} = -i, \quad w'(-1) = \frac{1}{2\sqrt{z}} \bigg|_{z=-1} = \frac{1}{2\sqrt{1}} e^{i \frac{\pi}{2}} = \frac{i}{2}.$$

14. 注意到 $\sqrt{(1-z^2)(1-k^2z^2)}$ 可分解成:

$$\sqrt{|(1-z^2)(1-k^2z^2)|} \exp \left(i \frac{\text{Arg}(1-z) + \text{Arg}(1+z) + \text{Arg}(1-kz) + \text{Arg}(1+kz)}{2} \right),$$

因此在 $z_1 = 1, z_2 = -1, z_3 = \frac{1}{k}, z_4 = -\frac{1}{k}$ 点附近绕行一圈, 函数值均会发生改变 (具体为 **其一辐角增加 2π**), 故 z_1, z_2, z_3, z_4 均为支点, 算上 $z_5 = \infty$, 共 5 个支点.

现在全平面作一闭合回路, 则其 z_1, z_3 与 z_2, z_4 必然**同时**被包含或不被包含, 因此辐角会增加 $0, 4\pi$ 或 8π , 函数辐角增加 $0, 2\pi$ 或 4π , 每个分支内函数值保持不变, 即为单值解析分支.

15. (与上一问一样, 待完成)

16. 含 e^z 形式的函数极限, **通常取 $z = \text{Re} z = x$ 并分别令 $x \rightarrow \pm\infty$ 加以反证**¹.

¹必要时还可令 $z = iy$ 进一步反证

(1) 令 $z = \operatorname{Re} z = x$, 则 $x \rightarrow +\infty$ 时, $\frac{z}{e^z} = \frac{x}{e^x} \rightarrow 0$; $x \rightarrow -\infty$ 时, $\frac{z}{e^z} = \frac{x}{e^x} \rightarrow -\infty$. 两者不相等, 故极限不存在.

(2) 令 $w = \frac{1}{z}$, 则 $z \rightarrow 0$ 时 $w \rightarrow \infty$, 且 $\sin w = \frac{e^{iw} - e^{-iw}}{2i}$. 令 $w = \operatorname{Re} w = u$, 则 $u \rightarrow \infty$ 时,

$$\frac{\sin w}{w} \rightarrow 0.$$

再令 $w = i\operatorname{Im} w = iv$, 则 $v \rightarrow \infty$ 时,

$$\frac{\sin w}{w} = \frac{e^{iw} - e^{-iw}}{2iw} = \frac{e^v - e^{-v}}{2v} \rightarrow +\infty.$$

可见两者不相等, 原极限不存在.

(3) 令 $z = \operatorname{Re} z = x$, 则 $x \rightarrow 1^+$ 时, $\frac{ze^{1/(z-1)}}{e^z - 1} = \frac{xe^{1/(x-1)}}{e^x - 1} \rightarrow \frac{+\infty}{e-1} = +\infty$; $x \rightarrow 1^-$ 时, $\frac{ze^{1/(z-1)}}{e^z - 1} \rightarrow \frac{-\infty}{e-1} = -\infty$. 两者不相等, 故极限不存在.

17. 取 $y = kx$, 其中 k 可从 $-\infty$ 取到 $+\infty$, 那么 $z = x + iy = (1 + ik)x$, 则

$$|z + e^z| \geq ||e^z| - |z|| = |e^x - \sqrt{1+k^2}|x||$$

对任意取定的 $|k| < +\infty$, 当 $x \rightarrow +\infty$ 时, $|z + e^z| \geq e^x - \sqrt{1+k^2}x$, 此时 $|z + e^z| \rightarrow +\infty$; 当 $x \rightarrow -\infty$ 时, $|z + e^z| \geq -\sqrt{1+k^2}x - e^x$, 即 $|z + e^z| \rightarrow +\infty + 0 = +\infty$.

特别地, 当 $k = \infty$ 时, $x = 0$, 代入得

$$|z + e^z| = |iy + e^{iy}| \geq |y - 1|,$$

因此 $y \rightarrow \infty$ 时, $|z + e^z| \rightarrow +\infty$.

综上, $\lim_{z \rightarrow \infty} (z + e^z) = \infty$.

18. (1) $\sin z = \frac{e^{iz} - e^{-iz}}{2i} = 2 \implies (e^{iz})^2 - 4ie^{iz} - 1 = 0 \implies e^{iz} = (2 \pm \sqrt{3})i$, 因此

$$\begin{aligned} z &= -i\operatorname{Ln}[(2 \pm \sqrt{3})i] = -i[\ln(2 \pm \sqrt{3}) + i\frac{\pi}{2} + 2k\pi i] \\ &= \left(2k + \frac{1}{2}\right)\pi - i\ln(2 \pm \sqrt{3}) \quad (k \in \mathbb{Z}). \end{aligned}$$

(2) $\cosh z = \frac{e^z + e^{-z}}{2} = 0 \implies e^{2z} = -1 = e^{i\pi} \implies z = \left(k + \frac{1}{2}\right)\pi i \quad (k \in \mathbb{Z})$.

(3) $e^z = A \implies z = \operatorname{Ln} A = \ln |A| + i(\arg A + 2k\pi) \quad (k \in \mathbb{Z})$.

19. (1) 奇点: $e^z + 1 = 0 \implies z = (2k + 1)\pi i \quad (k \in \mathbb{Z})$, 即解析区域为 $D = \{z \in \mathbb{C} | z \neq (2k + 1)\pi i, k \in \mathbb{Z}\}$, 该函数微商

$$\frac{d}{dz} \left(\frac{1}{1 + e^z} \right) = -\frac{e^z}{(1 + e^z)^2} \quad (z \in D).$$

(2) 奇点: $\sin z = \frac{e^{iz} - e^{-iz}}{2i} = 2 \implies z = \operatorname{Ln}[(2 \pm \sqrt{3})i] = \ln(2 \pm \sqrt{3}) + i\left(2k + \frac{1}{2}\right)\pi$,
即解析区域为

$$D = \{z \in \mathbb{C} | z \neq \ln(2 \pm \sqrt{3}) + i\left(2k + \frac{1}{2}\right)\pi, k \in \mathbb{Z}\}.$$

该函数微商

$$\frac{d}{dz} \left(\frac{1}{\sin z - 2} \right) = -\frac{\cos z}{(\sin z - 2)^2} \quad (z \in D).$$

(3) 奇点: $z - 1 = 0$, 即解析区域 $D = \{z \in \mathbb{C} | z \neq 1\}$, 该函数微商

$$\frac{d}{dz} \left(ze^{\frac{1}{z-1}} \right) = \left[1 - \frac{z}{(z-1)^2} \right] \exp \left(\frac{1}{z-1} \right).$$

20. (1) 由定义,

$$\begin{aligned} \cos(z_1 + z_2) &= \frac{e^{i(z_1+z_2)} + e^{-i(z_1+z_2)}}{2i} \\ &= \frac{(e^{iz_1} + e^{-iz_1})(e^{iz_2} + e^{-iz_2}) + (e^{iz_1} - e^{-iz_1})(e^{iz_2} - e^{-iz_2})}{4} \\ &= \frac{e^{iz_1} + e^{-iz_1}}{2} \cdot \frac{e^{iz_2} + e^{-iz_2}}{2} - \frac{e^{iz_1} - e^{-iz_1}}{2i} \cdot \frac{e^{iz_2} - e^{-iz_2}}{2i} \\ &= \cos z_1 \cos z_2 - \sin z_1 \sin z_2. \end{aligned}$$

(2) 由定义,

$$\begin{aligned} \sinh(z_1 + z_2) &= \frac{e^{z_1+z_2} - e^{-(z_1+z_2)}}{2} \\ &= \frac{e^{z_1} - e^{-z_1}}{2} \cdot \frac{e^{z_2} + e^{-z_2}}{2} + \frac{e^{z_1} + e^{-z_1}}{2} \cdot \frac{e^{z_2} - e^{-z_2}}{2} \\ &= \sinh z_1 \cosh z_2 + \cosh z_1 \sinh z_2. \end{aligned}$$

(3) 记 $w = \operatorname{Arccos} z$, 则 $z = \cos w = \frac{e^{iw} + e^{-iw}}{2} \implies (e^{iw})^2 - 2ze^{iw} + 1 = 0$, 解得²

$$w = \operatorname{Arccos} z = -i\operatorname{Ln}(z + \sqrt{z^2 - 1}).$$

21. 记 $z = x + iy$, $x, y \in \mathbb{R}$, 则

$$\sin z = \sin(x + iy) = \sin x \cos(iy) + \cos x \sin(iy) = \sin x \cosh y + i \cos x \sinh y,$$

因此 $\operatorname{Re}(\sin z) = \sin x \cosh y$, $\operatorname{Im}(\sin z) = \cos x \sinh y$,

$$\begin{aligned} |z| &= \sqrt{(\sin x \cosh y)^2 + (\cos x \sinh y)^2} = \sqrt{\sin^2 x (1 + \sinh^2 y) + \cos^2 x \sinh^2 y} \\ &= \sqrt{\sin^2 x + \sinh^2 y}. \end{aligned}$$

²这里根式函数默认多值, 因此无需“±”号

22. 记 $z = x + iy$, 则

$$\cos z = \cos(x + iy) = \cos x \cos(iy) - \sin x \sin(iy) = \cos x \cosh y - i \sin x \sinh y,$$

因此当 $\cos z \in \mathbb{R}$ 时, 其虚部

$$\operatorname{Im}(\cos z) = -\sin x \sinh y = 0 \implies \sin x = 0 \quad \text{或} \quad \sinh y = 0,$$

即 $x = \operatorname{Re} z = n\pi$ ($n \in \mathbb{Z}$) 或 $y = 0$.

23. (1) $\operatorname{Ln}(-1) = \ln 1 + i(\pi + 2k\pi) = (2k+1)\pi i$ ($k \in \mathbb{Z}$), $\ln(-1) = \pi i$;

$$(2) \operatorname{Lni} = \ln 1 + i\left(\frac{\pi}{2} + 2k\pi\right) = \left(2k + \frac{1}{2}\right)\pi i \quad (k \in \mathbb{Z}), \quad \ln i = \frac{\pi}{2}i;$$

$$(3) 1^{\sqrt{2}} = e^{\sqrt{2}\operatorname{Ln}1} = e^{2\sqrt{2}k\pi i}, \quad (-2)^{\sqrt{2}} = e^{\sqrt{2}\operatorname{Ln}(-2)} = e^{\sqrt{2}[\ln 2 + (2k+1)\pi i]},$$

$$2^i = e^{i\operatorname{Ln}2} = e^{-2k\pi + i\ln 2},$$

$$(3 - 4i)^{1+i} = \exp\left[(\ln 5 - 2k\pi + \arctan \frac{4}{3}) + i(\ln 5 + 2k\pi - \arctan \frac{4}{3})\right] \quad (k \in \mathbb{Z}).$$

$$(4) \cos(2 + i) = \cos 2 \cosh 1 - i \sin 2 \sinh 1, \quad \sin 2i = i \sinh 2,$$

$$\cot\left(\frac{\pi}{4} - i \ln 2\right) = \frac{\cos(i \ln 2) + \sin(i \ln 2)}{\cos(i \ln 2) - \sin(i \ln 2)} = \frac{\cosh(\ln 2) + i \sinh(\ln 2)}{\cosh(\ln 2) - i \sinh(\ln 2)} = \frac{8 + 15i}{17},$$

$$\coth(2 + i) = \frac{\cosh 2 \cos 1 + i \sinh 2 \sin 1}{\sinh 2 \cos 1 + i \cosh 2 \sin 1} = \frac{\sinh 2 \cosh 2 - i \sin 1 \cos 1}{\sin^2 1 + \sinh^2 2} = \frac{\sinh 4 - i \sin 2}{\cosh 4 - \cos 2}.$$

$$(5) \text{ 记 } w = \operatorname{Arcsin} i, \text{ 则 } i = \sin w = \frac{e^{iw} - e^{-iw}}{2i} \implies (e^{iw})^2 + 2e^{iw} - 1 = 0 \implies e^{iw} = -1 \pm \sqrt{2}, \text{ 即}$$

$$w_1 = -i\operatorname{Ln}(\sqrt{2} - 1) = 2k\pi - i \ln(\sqrt{2} - 1) \quad (k \in \mathbb{Z}),$$

$$w_2 = -i\operatorname{Ln}(-\sqrt{2} - 1) = (2k+1)\pi - i \ln(\sqrt{2} + 1) \quad (k \in \mathbb{Z}).$$

后同, 故略去.

24. 注意到 $(a^b)^c =$

2 解析函数的积分表示

1. C 上 z 的参数方程为 $z = 2e^{i\theta}$,

(1) θ 从 π 到 0, 因此

$$\begin{aligned}\int_C \frac{2z-3}{z} dz &= \int_{\pi}^0 \left(2 - \frac{3}{2}e^{-i\theta}\right) \cdot 2ie^{i\theta} d\theta = -i \int_0^{\pi} (4e^{i\theta} - 3) d\theta \\ &= -i \left(-4ie^{i\theta} - 3\theta\right) \Big|_0^{\pi} = 8 + 3\pi i.\end{aligned}$$

(2) θ 从 π 到 2π , 因此

$$\int_C \frac{2z-3}{z} dz = i \int_0^{\pi} (4e^{i\theta} - 3) d\theta = i \left(-4ie^{i\theta} - 3\theta\right) \Big|_{\pi}^{2\pi} = 8 - 3\pi i.$$

(3) θ 从 0 到 2π , 因此

$$\int_C \frac{2z-3}{z} dz = i \left(-4ie^{i\theta} - 3\theta\right) \Big|_0^{2\pi} = -6\pi i.$$

2. (1) 参数方程: $z = iy$ ($-1 \leq y \leq 1$), 因此

$$\int_{-i}^i |z| dz = \int_{-1}^1 |y| \cdot i dy = 2i \int_0^1 y dy = i.$$

(2) 参数方程: $z = e^{i\theta}$, θ 从 $\frac{3\pi}{2}$ 到 $\frac{\pi}{2}$, 因此

$$\int_{-i}^i |z| dz = \int_{\frac{3\pi}{2}}^{\frac{\pi}{2}} 1 \cdot ie^{i\theta} d\theta = 2i.$$

(3) 参数方程: $z = e^{i\theta}$, θ 从 $-\frac{\pi}{2}$ 到 $\frac{\pi}{2}$, 因此

$$\int_{-i}^i |z| dz = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} ie^{i\theta} d\theta = 2i.$$

3. (1) 代入 $x = 0$, 得

$$\left| \int_{-i}^i (x^2 + iy^2) dz \right| \leq \int_{-1}^1 |iy^2| |dz| \leq \int_{-1}^1 1 dy = 2.$$

(2) 注意到 $|x^2 + iy^2| = \sqrt{(x^2 + y^2)^2 - 2x^2y^2} \leq x^2 + y^2 = |z|^2 = 1$, 则

$$\left| \int_{-i}^i (x^2 + iy^2) dz \right| \leq \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1 |dz| = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta = \pi.$$

4. z 仅在 x 上变动, 虚部恒为 1, 因此

$$\left| \int_i^{2+i} \frac{dz}{z^2} \right| \leq \int_0^2 \frac{|dz|}{|z|^2} = \int_0^2 \frac{dx}{x^2 + 1} \leq \int_0^2 \frac{dx}{0 + 1} = 2.$$

5. 由于奇点 $z = -2$ 位于闭路 $|z| = 1$ 围成的闭区域外, 且 $\frac{1}{z+2}$ 在此区域内解析, 故由柯西积分定理,

$$\int_{|z|=1} \frac{1}{z+2} dz = 0.$$

注意到 z 在此闭路上参数方程为 $z = e^{i\theta}$, θ 从 0 到 2π , 因此

$$\int_{|z|=1} \frac{1}{z+2} dz = \int_0^{2\pi} \frac{ie^{i\theta} d\theta}{2 + e^{i\theta}} = i \int_0^{2\pi} \frac{2e^{i\theta} + 1}{5 + 4\cos\theta} d\theta.$$

代入 $\cos\theta = \cos(2\pi - \theta)$, 则有

$$\int_0^\pi \frac{1 + 2\cos\theta}{5 + 4\cos\theta} d\theta = \frac{1}{2} \int_0^{2\pi} \frac{1 + 2\cos\theta}{5 + 4\cos\theta} d\theta = \frac{1}{2} \operatorname{Im} \left(\int_{|z|=1} \frac{1}{z+2} dz \right) = 0.$$

6. (1) 原式 $= 2 \sin \frac{z}{2} \Big|_0^{\pi+2i} = 2 \cos i = 2 \cosh 1.$

(2) 勘误: 依答案此题上限应改为 “1” 而非 “i”, 但此处按原题干进行计算.

$$\text{原式} = z + iz^4 \Big|_{-i}^i = 2i.$$

(3) 原式 $= -e^{-z} \Big|_{-\pi i}^0 = -2.$

7. 构造 $\int_{C_R} \frac{dz}{z} = i\alpha$. 由题设条件, $\lim_{z \rightarrow \infty} zf(z) = A \implies$ 对 $\forall \varepsilon > 0$, 总 $\exists R_c(\varepsilon) > 0$, 当 $|z| > R_c$ 时, $|zf(z) - A| < \frac{\varepsilon}{\alpha}$, 故当 $R > R_c$ 时,

$$\begin{aligned} \left| \int_{C_R} f(z) dz - iA\alpha \right| &= \left| \int_{C_R} f(z) dz - A \int_{C_R} \frac{1}{z} dz \right| = \left| \int_{C_R} \frac{zf(z) - A}{z} dz \right| \\ &\leq \int_{C_R} \frac{|zf(z) - A|}{|z|} |dz| < \int_0^\alpha \frac{\varepsilon}{\alpha R} \cdot R d\theta = \varepsilon, \end{aligned}$$

因此 $\int_{C_R} f(z) dz = iA\alpha$.

8. 记 $f(z) = \frac{P(z)}{Q(z)}$, 那么 $\lim_{z \rightarrow \infty} zf(z) = 0$, 由上一题结论,

$$\lim_{R \rightarrow \infty} \int_{C_R} \frac{P(z)}{Q(z)} dz = \lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = i \cdot 0 \cdot 2\pi = 0.$$

9. 记 $f(z) = e^z$, 由柯西积分公式,

$$\int_{|z|=1} \frac{e^z}{z} dz = 2\pi i f(0) = 2\pi i \cdot 1 = 2\pi i.$$

设 $z = \cos\theta + i\sin\theta = e^{i\theta}$, 则

$$\int_{|z|=1} \frac{e^z}{z} dz = \int_0^{2\pi} \frac{e^{\cos\theta + i\sin\theta}}{e^{i\theta}} \cdot ie^{i\theta} d\theta = \int_0^{2\pi} e^{\cos\theta} [-\sin(\sin\theta) + i\cos(\sin\theta)] d\theta.$$

代入 $\cos\theta = \cos(2\pi - \theta)$, 得

$$\int_0^\pi e^{\cos\theta} \cos(\sin\theta) d\theta = \frac{1}{2} \int_0^{2\pi} e^{\cos\theta} \cos(\sin\theta) d\theta = \frac{1}{2} \operatorname{Im} \left(\int_{|z|=1} \frac{e^z}{z} dz \right) = 0.$$

10. $\frac{e^z}{1+z^2}$ 的奇点为 $z_1 = -i, z_2 = i$, 那么

(1) 记 $f(z) = \frac{e^z}{z+i}$, 由柯西积分公式,

$$\int_C \frac{e^z}{1+z^2} dz = 2\pi i f(i) = 2\pi i \cdot \frac{e^i}{2i} = \pi e^i.$$

(2) 记 $f(z) = \frac{e^z}{z-i}$, 由柯西积分公式,

$$\int_C \frac{e^z}{1+z^2} dz = 2\pi i f(-i) = 2\pi i \cdot \frac{e^{-i}}{-2i} = -\pi e^{-i}.$$

(3) 将其因式分解, 结合柯西积分公式可得

$$\begin{aligned} \int_C \frac{e^z}{1+z^2} dz &= \frac{1}{2i} \int_C \left(\frac{e^z}{z-i} - \frac{e^z}{z+i} \right) dz = 2\pi i \cdot \frac{1}{2i} (e^i - e^{-i}) \\ &= 2\pi i \sin 1. \end{aligned}$$

另解 利用柯西积分定理, 取 $z = i, -i$ 附近极小且路径沿正向的圆周, 则

$$\begin{aligned} \int_{|z|=2} \frac{dz}{1+z^2} &= \int_{|z-i| \leq \varepsilon_1} \frac{dz}{1+z^2} + \int_{|z+i| \leq \varepsilon_2} \frac{dz}{1+z^2} = \pi e^i - \pi e^{-i} \\ &= 2\pi i \sin 1. \end{aligned}$$

11. 分如下两种情况讨论:

(1) $r < 1$ 时, 函数只有奇点 $z = 0$, 则记 $f(z) = \frac{1}{(z+1)(z-1)}$,

$$\int_{|z|=r} \frac{dz}{z^2(z+1)(z-1)} = 2\pi i f'(0) = 2\pi i \cdot \frac{-2z}{(z^2-1)^2} \Big|_{z=0} = 0.$$

(2) $r > 1$ 时, 函数有奇点 $z = 0, 1, -1$, 则将其因式分解, 得

$$\begin{aligned} \int_{|z|=r} \frac{dz}{z^2(z+1)(z-1)} &= \frac{1}{2} \int_{|z|=r} \left(\frac{1}{z-1} - \frac{2}{z^2} - \frac{1}{z+1} \right) dz \\ &= \frac{1}{2} \cdot 2\pi i (1 - 0 - 1) = 0. \end{aligned}$$

另解 由柯西积分定理, 取 $z = 0, 1, -1$ 附近充分小且路径沿正向的圆周, 则

$$\begin{aligned} \int_{|z|=r} \frac{dz}{z^2(z+1)(z-1)} &= \sum_{k=1}^3 \int_{|z-k+2| \leq \varepsilon_k} \frac{dz}{z^2(z+1)(z-1)} \\ &= 2\pi i \cdot \left(-\frac{1}{2} - 0 + \frac{1}{2} \right) = 0. \end{aligned}$$

综上, 该积分值为零.

12. (1) 奇点 $z = -i$, 记 $f(z) = \frac{z}{9-z^2}$, 则

$$\int_C \frac{z dz}{(9-z^2)(z+i)} = 2\pi i f(-i) = 2\pi i \cdot \frac{-i}{10} = \frac{\pi}{5}.$$

(2) 奇点 $= -i, 3, -3$, 则

$$\begin{aligned}\int_C \frac{z dz}{(9-z^2)(z+i)} &= \frac{1}{6} \int_C \left(\frac{3}{5} \frac{1}{z+i} - \frac{1}{3-i} \frac{1}{z+3} - \frac{1}{3+i} \frac{1}{z-3} \right) dz \\ &= \frac{1}{6} \cdot 2\pi i \left(\frac{3}{5} \cdot 1 - \frac{1}{3-i} \cdot 1 - \frac{1}{3+i} \cdot 1 \right) = \frac{\pi}{3} i \cdot 0 = 0.\end{aligned}$$

另解 由柯西积分定理, 取 $z = -i, 3, -3$ 附近充分小且路径沿正向的圆周, 则

$$\int_C \frac{z dz}{(9-z^2)(z+i)} = 2\pi i \cdot \left[-\frac{i}{10} - \frac{1}{2(3+i)} - \frac{1}{2(-3+i)} \right] = 2\pi i \cdot \left(-\frac{i}{10} + \frac{i}{10} \right) = 0.$$

13. (1) $z_0 = 1$ 时, 其为 C 所包围的闭域内的奇点, 由柯西积分公式,

$$g(1) = 2\pi i \cdot (2z^2 - z + 1) \Big|_{z=1} = 4\pi i.$$

(2) $z_0 > 2$ 时, C 所包围的闭域内无奇点且被积函数在此闭域内解析, 由柯西积分定理, $g(z_0) \equiv 0$.

14. 记 $f(z) = \frac{z^2}{(z+i)^2}$, 由柯西积分公式,

$$\int_C \frac{z^2 dz}{(1+z^2)^2} = \int_C \frac{f(z)}{(z-i)^2} = 2\pi i f'(i) = 2\pi i \cdot \frac{2iz}{(z+i)^3} \Big|_{z=i} = \frac{\pi}{2}.$$

15. 注意到 $\text{Ln } p(z) = \sum_{i=1}^n \text{Ln}(z - a_i)$, 因此

$$\frac{p'(z)}{p(z)} = \frac{d}{dz} (\text{Ln } p(z)) = \sum_{i=1}^n \frac{1}{z - a_i}.$$

设 $p(z)$ 有 $a_{k_1}, a_{k_2}, \dots, a_{k_m}$ ($1 \leq k_1 \leq k_2 \leq \dots \leq k_m \leq n$) 这 m 个零点, 则

$$\frac{1}{2\pi i} \int_C \frac{1}{z - a_j} dz = \begin{cases} 1, & j \in \{k_1, k_2, \dots, k_m\}, \\ 0, & j \in \{1, 2, \dots, n\} - \{k_1, k_2, \dots, k_m\}. \end{cases}$$

因此 $\frac{1}{2\pi i} \int_C \frac{p'(z)}{p(z)} dz = 1 + 1 + \dots + 1 = m$, 此即多项式函数 $p(z)$ 的零点个数.

16. 假设存在这样的 $f(z)$, 当其在闭圆 $|z| \leq 1$ 内解析时, 以 C 为绕边界正向路径, 由柯西积分定理,

$$\int_C f(z) dz = 0.$$

又其在 C 上的值为 $f(z) = \frac{1}{z} = e^{-i\theta}$, 因此

$$\int_C f(z) dz = \int_0^{2\pi} e^{-i\theta} \cdot i e^{i\theta} d\theta = i \int_0^{2\pi} d\theta = 2\pi i \neq 0.$$

两个积分值不同, 矛盾, 故假设不成立, 这样的 $f(z)$ 并不存在.

17. 分两种情况讨论:

(1) 当 $z \in D$ 时, $\lim_{z \rightarrow \infty} f(z) = A \implies$ 对 $\forall \varepsilon > 0, z \in \mathbb{C}$, 总 $\exists R \in \mathbb{R}^+$, 当 $|\zeta| > R$ 时, 满足 $|f(\zeta) - A| < \varepsilon - |f(z) - A|$. 现作以点 z 为圆心、充分大的 R 为半径³的正向圆

³ Γ 围成的闭域包含 C 围成的闭域

周 Γ , 记 $L = \Gamma + C^-$, 由柯西积分公式,

$$\frac{1}{2\pi i} \int_L \frac{f(\zeta)}{\zeta - z} d\zeta$$

18. 当 $f(z)$ 在 D 内无零点时, 函数 $g(z) = \frac{1}{f(z)}$ 在 D 内同样解析且不恒为常数, 由**最大模原理**, $|g(z)|$ 的最大值不可能在 D 内取到, 即 $|f(z)|$ 的最小值不可能在 D 内取到.

19. **反证法**. 假设 $f(z)$ 在 D 内无零点, 同时 $f(z)$ 在 $|z| \leq a$ 内解析 (必然连续), 由上一问, $|f(z)|$ 的最小值只可能在边界上取到, 但对任一边界上的点, 都有 $|f(z)| > m > |f(0)|$, 两者矛盾. 故假设不成立, $f(z)$ 在 D 内至少有一个零点.

20. 对 $\forall R > 0$, $f(z) = \sum_{k=0}^n a_k z^k$ 总在 $|z| \leq R$ 内解析. 当 $|z| = R$ 时, 记 $M = \max\{a_n\}$, 则

$$\begin{aligned} |f(z)| \Big|_{|z|=R} &\geq |a_n| |z|^n - \sum_{k=0}^{n-1} |a_k| |z|^k = M \left(R^n - \frac{R^n - 1}{R - 1} \right) \\ &= \frac{R^n(R-2) + 1}{R-1} M > \frac{(R-2) + 1}{R-1} M = M. \end{aligned}$$

又因为 $|f(0)| = |a_0| \leq M$, 故 $f(z)$ 在 $|z| \leq R$ 内至少有一个零点. 考虑到 R 任取, 故 $f(z) = 0$ 在全平面上至少有一个零点.

3 调和函数

1. 由题意,

$$\frac{\partial^2 u}{\partial x^2} = 6ax + 2by, \quad \frac{\partial^2 u}{\partial y^2} = 2cx + 6dy,$$

由于 u 是调和函数, 故

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2(3a + c)x + 2(b + 3d)y \equiv 0,$$

即满足关系 $c = -3a, b = -3d$.

2. 设 $f(z) = u(x, y) + iv(x, y)$. 由于 $f(z)$ 解析, 故 $u(x, y), v(x, y)$ 均为调和函数, 那么

(1) 记 $g(x, y) = \ln |f(z)| = \ln \sqrt{u^2 + v^2} = \frac{1}{2} \ln(u^2 + v^2)$, 并改记 $x = x_1, y = x_2$, 则

$$\begin{aligned} \Delta g(x, y) &= \frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} = \nabla \cdot \left[\frac{1}{u^2 + v^2} \sum_{i=1}^2 \left(u \frac{\partial u}{\partial x_i} + v \frac{\partial v}{\partial x_i} \right) \hat{x}_i \right] \\ &= -\frac{2}{(u^2 + v^2)^2} \sum_{i=1}^2 \left(u \frac{\partial u}{\partial x_i} + v \frac{\partial v}{\partial x_i} \right)^2 \\ &\quad + \frac{1}{u^2 + v^2} \sum_{i=1}^2 \left[\left(\frac{\partial u}{\partial x_i} \right)^2 + \left(\frac{\partial v}{\partial x_i} \right)^2 + u \frac{\partial^2 u}{\partial x_i^2} + v \frac{\partial^2 v}{\partial x_i^2} \right] \\ &= \frac{(v^2 - u^2)|\nabla u|^2 + (u^2 - v^2)|\nabla v|^2 - 4uv(\nabla u \cdot \nabla v)}{(u^2 + v^2)^2} + \frac{u^2 \Delta u + v^2 \Delta v}{u^2 + v^2}. \end{aligned}$$

利用 C-R 方程可得

$$\begin{aligned} |\nabla u|^2 &= \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 = \left(\frac{\partial v}{\partial y} \right)^2 + \left(-\frac{\partial v}{\partial x} \right)^2 \\ &= |\nabla v|^2, \\ \nabla u \cdot \nabla v &= \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} \left(-\frac{\partial u}{\partial y} \right) + \frac{\partial u}{\partial y} \frac{\partial u}{\partial x} \\ &= 0, \end{aligned}$$

并代入 $\Delta u = \Delta v = 0$, 可得

$$\begin{aligned} \Delta g(x, y) &= \frac{[(v^2 - u^2) + (u^2 - v^2)]|\nabla u|^2 - 4uv \cdot 0}{(u^2 + v^2)^2} + \frac{0 + 0}{u^2 + v^2} \\ &= 0. \end{aligned}$$

因此 $g(x, y) = \ln |f(z)|$ 为调和函数.

(2) 直接验证:

$$\begin{aligned} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |f(z)|^2 &= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) (u^2 + v^2) = 2(|\nabla u|^2 + |\nabla v|^2 + u\Delta u + v\Delta v) \\ &= 2(|\nabla u|^2 + |\nabla v|^2) = 4 \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(-\frac{\partial u}{\partial y} \right)^2 \right] \\ &= 4|f'(z)|^2. \end{aligned}$$

3. 由题意, $\Delta u = 0$, 则

(1) 对于 u^2 ,

$$\Delta(u^2) = 2\nabla \cdot \left(u \frac{\partial u}{\partial x} \hat{x} + u \frac{\partial u}{\partial y} \hat{y} \right) = 2(|\nabla u|^2 + u\Delta u) = 2|\nabla u|^2 \geq 0.$$

由于 u 不恒为常数, 故 ∇u 不恒为 $\mathbf{0}$, 即 Δu 不恒为 0, u^2 不是调和函数.

(2) 当 $f(u)$ 是调和函数时,

$$\begin{aligned} \Delta f(u) &= \nabla \cdot \left[f'(u) \left(\frac{\partial u}{\partial x} \hat{x} + \frac{\partial u}{\partial y} \hat{y} \right) \right] = f'(u)\Delta u + f''(u)|\nabla u|^2 \\ &= f''(u)|\nabla u|^2 \equiv 0, \end{aligned}$$

由上一问可知, $|\nabla u|$ 不恒为 0, 因此等式成立当且仅当 $f''(u) \equiv 0$, 积分得

$$f(u) = C_1 u + C_2,$$

其中 C_1, C_2 为由初始条件确定的常数, 即要求 $f(u)$ 是一次函数.

4. 设 $f(z) = u(x, y) + iv(x, y)$, 由题意, $f(z)$ 解析.

(1) 先验证 $u(x, y)$ 为调和函数:

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} (3x^2 - 12xy - 3y^2) = 6x - 12y, \\ \frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y} (-6x^2 - 6xy + 6y^2) = -6x + 12y, \end{aligned}$$

故 $\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, 即 u 为调和函数.

由 C-R 方程, 虚部各偏导

$$\begin{aligned} \frac{\partial v}{\partial x} &= -\frac{\partial u}{\partial y} = 6(x^2 + xy - y^2), \\ \frac{\partial v}{\partial y} &= \frac{\partial u}{\partial x} = 3(x^2 - 4xy - y^2), \end{aligned}$$

代入初始条件 $v(0, 0) = \text{Im } f(0) = 0$, 积分得

$$\begin{aligned} v(x, y) &= v(0, 0) + \int_{(0,0)}^{(x,y)} 6(s^2 + st - t^2) ds + 3(s^2 - 4st - t^2) dt \\ &= 6 \int_0^x (s^2 + st - t^2) \Big|_{t=0} ds + 3 \int_0^y (s^2 - 4st - t^2) \Big|_{s=x} dt \\ &= 2x^3 + 3x^2y - 6xy^2 - y^3, \end{aligned}$$

因此

$$\begin{aligned} f(z) &= u + iv = (x^3 - 6x^2y - 3xy^2 + 2y^3) + i(2x^3 + 3x^2y - 6xy^2 - y^3) \\ &= x^3(1 + 2i) + 3ix^2y(1 + 2i) - 3xy^2(1 + 2i) - iy^3(1 + 2i) \\ &= [x^3 + 3x^2(iy) + 3x(iy)^2 + (iy)^3](1 + 2i) = (x + iy)^3(1 + 2i) \\ &= (1 + 2i)z^3. \end{aligned}$$

(2) 先验证 $u(x, y)$ 为调和函数:

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} \{e^x[(x+1)\cos y - y\sin y]\} = e^x[(x+2)\cos y - y\sin y], \\ \frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y} \{-e^x[y\cos y + (x+1)\sin y]\} = -e^x[(x+2)\cos y - y\sin y],\end{aligned}$$

故 $\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, 即 u 为调和函数.

由 C-R 方程, 虚部各偏导

$$\begin{aligned}\frac{\partial v}{\partial x} &= -\frac{\partial u}{\partial y} = e^x[(x+1)\sin y + y\cos y], \\ \frac{\partial v}{\partial y} &= \frac{\partial u}{\partial x} = e^x[(x+1)\cos y - y\sin y],\end{aligned}$$

代入初始条件 $v(0, 0) = \operatorname{Im} f(0) = 0$, 积分得

$$\begin{aligned}v(x, y) &= v(0, 0) + \int_{(0,0)}^{(x,y)} e^s[(s+1)\sin t + t\cos t] ds + e^s[(s+1)\cos t - t\sin t] dt \\ &= \int_0^x e^s[(s+1)\sin t + t\cos t] \Big|_{t=0} ds + \int_0^y e^s[(s+1)\cos t - t\sin t] \Big|_{s=x} dt \\ &= 0 + e^x[(x+1)\sin y - y\cos y + \sin y] = e^x(x\sin y + y\cos y),\end{aligned}$$

因此

$$\begin{aligned}f(z) &= u + iv = e^x(x\cos y - y\sin y) + ie^x(x\sin y + y\cos y) \\ &= xe^x(\cos y + i\sin y) + iye^x(\cos y + i\sin y) = (x + iy)e^{x+iy} \\ &= ze^z.\end{aligned}$$

(3) 先验证 $v(x, y)$ 为调和函数:

$$\begin{aligned}\frac{\partial^2 v}{\partial x^2} &= \frac{\partial}{\partial x} \left\{ \frac{2(x+1)y}{[(x+1)^2 + y^2]^2} \right\} = \frac{2y[y^2 - 3(x+1)^2]}{[(x+1)^2 + y^2]^3}, \\ \frac{\partial^2 v}{\partial y^2} &= \frac{\partial}{\partial y} \left\{ -\frac{(x+1)^2 - y^2}{[(x+1)^2 + y^2]^2} \right\} = \frac{6(x+1)^2 y - 2y^3}{[(x+1)^2 + y^2]^3},\end{aligned}$$

故 $\Delta u = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$, 即 v 为调和函数.

由 C-R 方程, 实部各偏导

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} = -\frac{(x+1)^2 - y^2}{[(x+1)^2 + y^2]}, \\ \frac{\partial u}{\partial y} &= -\frac{\partial v}{\partial x} = -\frac{2(x+1)y}{[(x+1)^2 + y^2]},\end{aligned}$$

代入初始条件 $u(0,0) = \operatorname{Re} f(0) = 2$, 积分得

$$\begin{aligned}
 u(x,y) &= u(0,0) - \int_{(0,0)}^{(x,y)} \frac{(s+1)^2 - t^2}{[(s+1)^2 + t^2]} \mathrm{d}s + \frac{2(s+1)t}{[(s+1)^2 + t^2]} \mathrm{d}t \\
 &= 2 - \int_0^x \frac{(s+1)^2 - t^2}{[(s+1)^2 + t^2]^2} \Big|_{t=0} \mathrm{d}s - \int_0^y \frac{2(s+1)t}{[(s+1)^2 + t^2]^2} \Big|_{s=x} \mathrm{d}t \\
 &= 2 - \int_0^x \frac{\mathrm{d}s}{(s+1)^2} - 2(x+1) \int_0^y \frac{t \mathrm{d}t}{[(x+1)^2 + t^2]^2} \\
 &= 2 - \frac{x}{x+1} - \frac{1}{x+1} + \frac{x+1}{(x+1)^2 + y^2} = 1 + \frac{x+1}{(x+1)^2 + y^2}.
 \end{aligned}$$

因此

$$\begin{aligned}
 f(z) = u + \mathrm{i}v &= \left[1 + \frac{x+1}{(x+1)^2 + y^2} \right] - \mathrm{i} \frac{y}{(x+1)^2 + y^2} = 1 + \frac{(x+1) - \mathrm{i}y}{(x+1)^2 + y^2} \\
 &= 1 + \frac{1}{(x+1) + \mathrm{i}y} = 1 + \frac{1}{1+z}.
 \end{aligned}$$

5. 由于 $u(z)$ 在全平面有界且调和, 那么可确定唯一的解析函数 $f(z)$, 满足 $u(z) = \operatorname{Re} f(z)$. 利用指数函数性质, 构造函数 $\mathrm{e}^{f(z)}$, 由于 $u(z)$ 有界, 即 $\exists M > 0$, 使得 $|u(z)| \leq M$, 故

$$|\mathrm{e}^{f(z)}| = |\mathrm{e}^u| \cdot |\mathrm{e}^{\mathrm{i}v}| = \mathrm{e}^u \in [\mathrm{e}^{-M}, \mathrm{e}^M],$$

即 $\mathrm{e}^{f(z)}$ 在全平面同样有界. 由 Liouville 定理, $\mathrm{e}^{f(z)}$ 为常数, 即 $f(z)$ 为常数.

4 解析函数的级数展开

1. (1) 注意到

$$\left| \frac{z^n}{n^2} \right| = \frac{|z|^n}{n^2} = \frac{1}{n},$$

而数项级数 $\sum_{n=1}^{\infty} \frac{1}{n^2}$ 收敛, 故 $\sum_{n=1}^{\infty} \frac{z^n}{n^2}$ 绝对收敛.

(2) 由于 $|z^n| = |z|^n \equiv 1$, 故 $\lim_{n \rightarrow \infty} z_n$ 必定非 0, 因此 $\sum_{n=1}^{\infty} z^n$ 发散.

(3) 当 $z = 1$ 时, 级数 $\sum_{n=1}^{\infty} \frac{z^n}{n} = \sum_{n=1}^{\infty} \frac{1}{n}$ 发散; 当 $z = -1$ 时, 级数 $\sum_{n=1}^{\infty} \frac{z^n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$.
由 **Leibniz 判别法**, 该交错级数收敛.

2. (1) 注意到 $|z| \leq 1$ 时,

$$\left| \frac{z^n}{n^2} \right| = \frac{|z|^n}{n^2} \leq \frac{1}{n^2},$$

而数项级数 $\sum_{n=1}^{\infty} \frac{1}{n^2}$ 收敛, 由 Weierstrass 判别法, 原级数在 $|z| \leq 1$ 上绝对一致收敛.

(2) i. 当 $|z| \leq r$ ($r < 1$) 时,

$$|z^n| = |z|^n \leq r^n,$$

而数项级数 $\sum_{n=1}^{\infty} r^n = \frac{r}{1-r}$ ($r < 1$) 收敛, 由 Weierstrass 判别法, 原级数绝对一致收敛.

ii. 当 $|z| < 1$ 时, 考虑该级数的部分和:

$$S_n(z) = \sum_{k=1}^n z^k = \frac{1 - z^{n+1}}{1 - z} z.$$

可见 $\{S_n(z)\}$ **逐点收敛** 于 $f(z) = \lim_{n \rightarrow \infty} S_n = \frac{z}{1-z}$.

假定 $\{S_n(z)\}$ 一致收敛于 $f(z)$, 即对于 $\forall \varepsilon > 0$, 总 $\exists N(\varepsilon) \in \mathbb{N}^*$, 使得当 $n > N$ 时, 满足

$$|S_n(z) - f(z)| = \left| \frac{1 - z^{n+1}}{1 - z} z - \frac{z}{1 - z} \right| = \frac{|z|^{n+1}}{|1 - z|} < \varepsilon.$$

另一方面, 可取 $z = \frac{1}{\sqrt[n+1]{2}} \in \mathbb{R}^+$, 使得

$$|S_n(z) - f(z)| = \frac{|z|^{n+1}}{|1 - z|} > |z|^{n+1} = \frac{1}{2}.$$

即所选择的 ε 不能任意小, 两者相矛盾, 因此假设不成立, $\{S_n(z)\}$ 即级数 $\sum_{n=1}^{\infty} z^n$ 在 $|z| < 1$ 上不一致收敛.

3. (1) 函数的唯一奇点为 $z_0 = 1$, 因此收敛域为 $|z| < 1$.

$$\frac{1}{1-z} + e^z = \sum_{n=0}^{\infty} z^n + \sum_{n=0}^{\infty} \frac{z^n}{n!} = \sum_{n=0}^{\infty} \left(1 + \frac{1}{n!}\right) z^n, \quad |z| < 1.$$

(2) 函数无奇点, 收敛域为全平面, 代入 $\cos z$ 的 Taylor 展开式得

$$\begin{aligned} (1-z+z^2)\cos z &= (1-z+z^2) \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!} = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n} - z^{2n+1} + z^{2n+2}}{(2n)!} \\ &= 1 - z + \sum_{n=1}^{\infty} \left[\frac{(-1)^n}{(2n)!} + \frac{(-1)^{n-1}}{(2n-2)!} \right] z^{2n} - \frac{(-1)^n}{(2n)!} z^{2n+1} \\ &= 1 - z + \sum_{n=1}^{\infty} (-1)^{n-1} \frac{4n^2 - 2n - 1}{(2n)!} z^{2n} - \frac{(-1)^n}{(2n)!} z^{2n+1} \\ &= 1 - z + \sum_{n=1}^{\infty} \left[(1-n-n^2) \cos \frac{n\pi}{2} - \sin \frac{n\pi}{2} \right] \frac{z^n}{n!} \\ &= - \sum_{n=0}^{\infty} \left[(n^2 - n - 1) \cos \frac{n\pi}{2} + \sin \frac{n\pi}{2} \right] \frac{z^n}{n!}, \quad |z| < \infty. \end{aligned}$$

(3) 函数无奇点, 收敛域为全平面, 直接展开:

$$e^{-z^2} = \sum_{n=0}^{\infty} \frac{(-z^2)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} z^{2n}, \quad |z| < \infty.$$

(4) 函数无奇点, 收敛域为全平面. 考虑 $e^z(\cos z \pm i \sin z) = e^{(1 \pm i)z}$, 得

$$\begin{aligned} e^z(\cos z + i \sin z) &= e^{(1+i)z} = \sum_{n=0}^{\infty} \frac{[(1+i)z]^n}{n!} = \sum_{n=0}^{\infty} \frac{(\sqrt{2})^n e^{i\frac{n\pi}{4}}}{n!} z^n, \\ e^z(\cos z - i \sin z) &= e^{(1-i)z} = \sum_{n=0}^{\infty} \frac{[(1-i)z]^n}{n!} = \sum_{n=0}^{\infty} \frac{(\sqrt{2})^n e^{-i\frac{n\pi}{4}}}{n!} z^n, \end{aligned}$$

两式相加除以 2 得

$$e^z \cos z = \sum_{n=0}^{\infty} (\sqrt{2})^n \frac{e^{i\frac{n\pi}{4}} + e^{-i\frac{n\pi}{4}}}{2} \frac{z^n}{n!} = \sum_{n=0}^{\infty} \frac{(\sqrt{2})^n}{n!} \cos \frac{n\pi}{4} z^n, \quad |z| < \infty.$$

(5) 函数奇点为 $z_1 = 1, z_2 = 2$, 因此收敛域为 $|z| < 1$, 分解后展开:

$$\frac{1}{z^2 - 3z + 2} = \frac{1}{1-z} - \frac{1}{2-z} = \sum_{n=0}^{\infty} z^n - \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{z}{2}\right)^n = \sum_{n=0}^{\infty} \left(1 - \frac{1}{2^{n+1}}\right) z^n, \quad |z| < 1.$$

(6) 函数无奇点, 收敛域为全平面, 利用倍角公式,

$$\begin{aligned} \sin^2 z &= \frac{1 - \cos 2z}{2} = \frac{1}{2} - \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (2z)^{2n} = \frac{1}{2} + \sum_{n=0}^{\infty} \frac{(-2)^{n-1}}{(2n)!} z^{2n} \\ &= \sum_{n=1}^{\infty} \frac{(-2)^{n-1}}{(2n)!} z^{2n}, \quad |z| < \infty. \end{aligned}$$

- (7) 函数奇点为所有使 $\cos z = 0$ 的点, 即 $e^{2iz} = -1$, 解得 $z = i\left(k + \frac{1}{2}\right)\pi$ ($k \in \mathbb{Z}$). 注意到 $|z| = \left|k + \frac{1}{2}\pi\right| \geq \frac{\pi}{2}$, 因此收敛域为 $|z| < \frac{\pi}{2}$. 设

$$\tan z = \sum_{n=0}^{\infty} a_n z^n, \quad |z| < \frac{\pi}{2},$$

由于题目只要求写出前四项⁴, 并且 $\cos z, \tan z$ 的 Taylor 级数在此收敛域内均一致收敛, 那么

$$\begin{aligned} \sin z &= \cos z \tan z = \left(1 - \frac{1}{2!}z^2 + \frac{1}{4!}z^4 + \cdots\right)(a_0 + a_1z + a_2z^2 + a_3z^3 + \cdots) \\ &= a_0 + a_1z + \left(a_2 - \frac{a_0}{2}\right)z^2 + \left(a_3 - \frac{a_1}{2}\right)z^3 + \cdots \\ &= z - \frac{1}{3!}z^3 + \frac{1}{5!}z^5 + \cdots, \end{aligned}$$

对应系数解得 $a_0 = a_2 = 0, a_1 = 1, a_3 = \frac{1}{3}$, 因此

$$\tan z = z + \frac{1}{3}z^3 + \cdots, \quad |z| < \frac{\pi}{2}.$$

- (8) 函数有唯一奇点 $z_0 = 1$, 因此收敛域为 $|z| < 1$. 注意到 $\frac{1}{1-z}$ 的 Taylor 级数一致收敛于其本身, 利用导数展开:

$$\frac{z}{(1-z)^2} = z \frac{d}{dz} \left(\frac{1}{1-z} \right) = z \frac{d}{dz} \left(\sum_{n=0}^{\infty} z^n \right) = z \sum_{n=0}^{\infty} \frac{dz^n}{dz} = \sum_{n=0}^{\infty} n z^n, \quad |z| < 1.$$

- (9) 函数无奇点, 收敛域为全平面, 注意到

$$e^{z^2} = \sum_{n=0}^{\infty} \frac{(z^2)^n}{n!} = \sum_{n=0}^{\infty} \frac{z^{2n}}{n!},$$

并且 e^{z^2} 的 Taylor 级数一致收敛于其本身, 故

$$\int_0^z e^{u^2} du = \int_0^z \sum_{n=0}^{\infty} \frac{u^{2n}}{n!} du = \sum_{n=0}^{\infty} \int_0^z \frac{u^{2n}}{n!} du = \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)n!}, \quad |z| < \infty.$$

- (10) 由于 $\sin z$ 在全平面无奇点, 且

$$\begin{aligned} \sin z &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+1}, \\ \frac{\sin z}{z} &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n}, \end{aligned}$$

可见 $\frac{\sin z}{z}$ 在全平面也解析, 收敛域为全平面, 因此

$$\begin{aligned} \int_0^z \frac{\sin u}{u} du &= \int_0^z \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} u^{2n} du = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \int_0^z u^{2n} du \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+2)! - (2n+1)!}, \quad |z| < \infty. \end{aligned}$$

⁴ 根据答案意思应该是展到第三阶即 z^0, z^1, z^2, z^3 这四项

4. 由题中所设,

$$\begin{aligned} 1 &= (1 - z - z^2) \sum_{n=0}^{\infty} C_n z^n = \sum_{n=0}^{\infty} C_n (z^n - z^{n+1} - z^{n+2}) \\ &= C_0 + (C_1 - C_0)z + \sum_{n=0}^{\infty} (C_{n+2} - C_{n+1} - C_n)z^{n+2}, \end{aligned}$$

对比系数可知, $C_0 = C_1 = 1$, $C_{n+2} - C_{n+1} - C_n = 0$ (恰为斐波那契数列), 因此前五项为

$$C_0 = C_1 = 1, \quad C_2 = C_0 + C_1 = 2, \quad C_3 = C_1 + C_2 = 3, \quad C_4 = C_2 + C_3 = 5,$$

写成展开式为

$$\frac{1}{1 - z - z^2} = 1 + z + 2z^2 + 3z^3 + 5z^4 + \cdots.$$

该递推的特征方程为 $\lambda^2 - \lambda - 1 = 0$, 解得特征根

$$\lambda_1 = \frac{1 + \sqrt{5}}{2} \in (1, \infty), \quad \lambda_2 = \frac{1 - \sqrt{5}}{2} \in (-1, 0),$$

则各系数形式为

$$C_n = A_1 \lambda_1^n + A_2 \lambda_2^n = A_1 \left(\frac{1 + \sqrt{5}}{2} \right)^n + A_2 \left(\frac{1 - \sqrt{5}}{2} \right)^n,$$

因此收敛半径

$$R = \lim_{n \rightarrow \infty} \left| \frac{C_n}{C_{n+1}} \right| = \lim_{n \rightarrow \infty} \frac{1}{|\lambda_1|} = \frac{\sqrt{5} - 1}{2}.$$

5. (1) 在其收敛域内, 题设级数 $\sum_{n=0}^{\infty} p_n(z) t^n$ 一致收敛于函数 $\frac{1}{\sqrt{1 - 2tz + t^2}}$, 故可以逐项求导:

$$\begin{aligned} \frac{d}{dt} \left(\frac{1}{\sqrt{1 - 2tz + t^2}} \right) &= \frac{d}{dt} \left(\sum_{n=0}^{\infty} p_n(z) t^n \right) = \sum_{n=0}^{\infty} \frac{d[p_n(z) t^n]}{dt} \\ &= \sum_{n=1}^{\infty} n p_n(z) t^{n-1} = \sum_{n=0}^{\infty} (n+1) p_{n+1}(z) t^n \\ &= -\frac{t - z}{(1 - 2tz + t^2)^{3/2}}, \end{aligned}$$

移项得

$$\begin{aligned} &(1 - 2tz + t^2) \sum_{n=0}^{\infty} (n+1) p_{n+1}(z) t^n \\ &= \sum_{n=0}^{\infty} (n+1) p_{n+1}(z) (t^n - 2zt^{n+1} + t^{n+2}) \\ &= p_1(z) + \sum_{n=0}^{\infty} [(n+2) p_{n+2}(z) - 2z(n+1) p_{n+1}(z) + n p_n(z)] t^{n+1}, \end{aligned}$$

又因为

$$\begin{aligned}(1-2tz+t^2)\sum_{n=0}^{\infty}(n+1)p_{n+1}(z)t^n &= -\frac{t-z}{\sqrt{1-2tz+t^2}} = -(t-z)\sum_{n=0}^{\infty}p_n(z)t^n \\ &= zp_0(z) + \sum_{n=0}^{\infty}[zp_{n+1}(z) - p_n(z)]t^{n+1},\end{aligned}$$

由Taylor 展开式的唯一性, 对比系数可知

$$(n+2)p_{n+2}(z) - 2z(n+1)p_{n+1}(z) + np_n(z)t^{n+1} = zp_{n+1}(z) - p_n(z) \quad (n \in \mathbb{N}),$$

整理可得

$$(n+1)p_{n+1}(z) - (2n+1)zp_n(z) + np_{n-1}(z) = 0 \quad (n \geq 1). \quad (1)$$

(2) 条件同上一问, 但此时变为对 z 求导:

$$\begin{aligned}\frac{d}{dz}\left(\frac{1}{\sqrt{1-2tz+t^2}}\right) &= \frac{d}{dz}\left(\sum_{n=0}^{\infty}p_n(z)t^n\right) = \sum_{n=0}^{\infty}\frac{d[p_n(z)t^n]}{dz} = \sum_{n=0}^{\infty}p'_n(z)t^n \\ &= \frac{t}{(1-2tz+t^2)^{3/2}},\end{aligned}$$

移项得

$$\begin{aligned}(1-2tz+t^2)\sum_{n=0}^{\infty}p'_n(z)t^n \\ &= \sum_{n=0}^{\infty}p'_n(z)(t^n - 2zt^{n+1} + t^{n+2}) \\ &= p'_0(z) + [p'_1(z) - 2zp'_0(z)]t + \sum_{n=0}^{\infty}[p'_{n+2}(z) - 2zp'_{n+1}(z) + p'_n(z)]t^{n+2},\end{aligned}$$

又因为

$$\begin{aligned}(1-2tz+t^2)\sum_{n=0}^{\infty}p'_n(z)t^n &= \frac{t}{\sqrt{1-2tz+t^2}} = t\sum_{n=0}^{\infty}p_n(z)t^n \\ &= p_0(z)t + \sum_{n=0}^{\infty}p_{n+1}(z)t^{n+2},\end{aligned}$$

由Taylor 展开式的唯一性, 对比系数可知

$$p'_{n+2}(z) - 2zp'_{n+1}(z) + p'_n(z) = p_{n+1}(z) \quad (n \in \mathbb{N}),$$

整理可得

$$p_n(z) = p'_{n+1}(z)2 - 2zp'_n(z) + p'_{n-1}(z) \quad (n \geq 1). \quad (2)$$

(3) 式(1) $\times 2$ 再对 z 求导得

$$2(2n+1)p_n(z) + 2(2n+1)zp'_n(z) = (2n+2)p'_{n+1}(z) + np'_{n-1}(z), \quad (3)$$

式(2) $\times (2n+1)$ - (3) 得

$$(2n+1)p_n(z) = p'_{n+1}(z) - p'_{n-1}(z) \quad (n \geq 1). \quad (4)$$

6. 构造复数 $z = ae^{i\theta}$, 那么 $|z| = |a| < 1$, 且

$$\begin{aligned}\frac{1}{1-z} &= \frac{1}{1-ae^{i\theta}} = \frac{1-ae^{-i\theta}}{(1-ae^{i\theta})(1-ae^{-i\theta})} \\ &= \frac{(1-a\cos\theta) + i(a\sin\theta)}{1-2a\cos\theta+a^2}.\end{aligned}$$

而 $\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$.

(1) 取其实部:

$$\begin{aligned}\frac{1-a\cos\theta}{1-2a\cos\theta+a^2} &= \operatorname{Re}\left(\frac{1}{1-z}\right) = \operatorname{Re}\left(\sum_{n=0}^{\infty} z^n\right) = \operatorname{Re}\left(\sum_{n=0}^{\infty} a^n e^{in\theta}\right) \\ &= \sum_{n=0}^{\infty} a^n \cos n\theta.\end{aligned}$$

(2) 取其虚部:

$$\begin{aligned}\frac{a\sin\theta}{1-2a\cos\theta+a^2} &= \operatorname{Im}\left(\frac{1}{1-z}\right) = \operatorname{Im}\left(\sum_{n=0}^{\infty} z^n\right) = \operatorname{Im}\left(\sum_{n=0}^{\infty} a^n e^{in\theta}\right) \\ &= \sum_{n=0}^{\infty} a^n \sin n\theta.\end{aligned}$$

(3) 对 $\frac{1}{1-z} = \frac{1}{1-ae^{i\theta}}$ 在 $(0, z)$ 上积分得

$$\operatorname{Ln}(1-z) = \int_0^z \sum_{n=0}^{\infty} u^n du = \sum_{n=0}^{\infty} \int_0^z u^n du = \sum_{n=0}^{\infty} \frac{z^{n+1}}{n+1}.$$

对 $\frac{1}{1-\bar{z}} = \frac{1}{1-ae^{-i\theta}}$ 在 $(0, \bar{z})$ 上积分得

$$\operatorname{Ln}(1-\bar{z}) = \int_0^{\bar{z}} \sum_{n=0}^{\infty} \bar{u}^n d\bar{u} = \sum_{n=0}^{\infty} \int_0^{\bar{z}} \bar{u}^n d\bar{u} = \sum_{n=0}^{\infty} \frac{\bar{z}^{n+1}}{n+1}.$$

两式相加得

$$\operatorname{Ln}(1-z)(1-\bar{z}) = \sum_{n=0}^{\infty} \frac{z^{n+1} + \bar{z}^{n+1}}{n+1} = \sum_{n=0}^{\infty} \frac{2a^{n+1} \cos(n+1)\theta}{n+1},$$

代入 $(1-z)(1-\bar{z}) = |1-z|^2 = 1-2a\cos\theta+a^2$, 整理得

$$\ln(1-2a\cos\theta+a^2) = 2 \sum_{n=1}^{\infty} \frac{a^n \cos n\theta}{n}.$$

7. 利用指数函数的 Taylor 展开,

$$\begin{aligned}|e^z - 1| &= \left| \sum_{n=1}^{\infty} \frac{z^n}{n!} \right| \leq \sum_{n=1}^{\infty} \frac{|z|^n}{n!} = e^{|z|} - 1 \\ &= |z| \sum_{n=1}^{\infty} \frac{|z|^{n-1}}{n!} = |z| \sum_{n=0}^{\infty} \frac{|z|^n}{(n+1)!} \leq |z| \sum_{n=0}^{\infty} \frac{|z|^n}{n!} = |z|e^{|z|},\end{aligned}$$

综上,

$$|e^z - 1| \leq e^{|z|} - 1 \leq |z|e^{|z|}.$$

8. 设 $f(z)$ 具有 m ($m \geq n$) 级零点, 则 $\exists f_1(z), \varphi_1(z)$ 在 z_0 点解析且非零, 且使得

$$f(z) = (z - z_0)^m f_1(z), \quad \varphi(z) = (z - z_0)^n \varphi_1(z),$$

那么

$$\begin{aligned} \lim_{z \rightarrow z_0} \frac{f(z)}{\varphi(z)} &= \lim_{z \rightarrow z_0} \frac{(z - z_0)^m f_1(z)}{(z - z_0)^n \varphi_1(z)} = \frac{f_1(z_0)}{\varphi_1(z_0)} \lim_{z \rightarrow z_0} (z - z_0)^{m-n} \\ &= \begin{cases} \frac{f_1(z_0)}{\varphi_1(z_0)}, & m = n, \\ 0, & m > n. \end{cases} \end{aligned}$$

当 $m \geq n$ 时, 由 Leibniz 公式,

$$\begin{aligned} \lim_{z \rightarrow z_0} f^{(n)}(z) &= \lim_{z \rightarrow z_0} \sum_{k=0}^n \binom{n}{k} f_1^{(k)}(z) \frac{m!}{(m-n+k)!} (z - z_0)^{m-n+k} \\ &= \frac{m!}{(m-n)!} f_1(z_0) \lim_{z \rightarrow z_0} (z - z_0)^{m-n} + 0 + 0 + \cdots + 0 \\ &= \begin{cases} f_1(z_0), & m = n, \\ 0, & m > n, \end{cases} \\ \lim_{z \rightarrow z_0} \varphi^{(n)}(z) &= \lim_{z \rightarrow z_0} \sum_{k=0}^n \binom{n}{k} \varphi_1^{(k)}(z) \frac{n!}{k!} (z - z_0)^k = \varphi_1(z_0), \end{aligned}$$

因此

$$\lim_{z \rightarrow z_0} \frac{f^{(n)}(z)}{\varphi_1^{(n)}(z)} = \begin{cases} \frac{f_1(z_0)}{\varphi_1(z_0)}, & m = n, \\ 0, & m > n \end{cases} = \lim_{z \rightarrow z_0} \frac{f(z)}{\varphi(z)}.$$

9. 由题意, 总 $\exists f_1(z), g_1(z)$ 在 z_0 点解析且非零, 则有

$$f(z) = (z - z_0)^m f_1(z), \quad g(z) = (z - z_0)^n g_1(z).$$

(1) 代入得

$$f(z)g(z) = (z - z_0)^{m+n} [f_1(z)g_1(z)].$$

可见 $f_1(z_0)g_1(z_0)$ 解析且非零, 故 $f(z)g(z)$ 具有 $(m+n)$ 阶零点.

(2) 代入得

$$\begin{aligned} f(z) + g(z) &= (z - z_0)^m f_1(z) + (z - z_0)^n g_1(z) \\ &= (z - z_0)^n [(z - z_0)^{m-n} f_1(z) + g_1(z)], \end{aligned}$$

注意到

$$[(z - z_0)^{m-n} f_1(z) + g_1(z)] \Big|_{z=z_0} = \begin{cases} f_1(z) + g_1(z), & m = n, \\ g_1(z), & m > n \end{cases} \neq 0,$$

因此当 $m > n$ 时, $f(z) + g(z)$ 具有 n 阶零点; 当 $m = n$ 时, 由于可能出现 $(z - z_0)[f_1(z_0) + g_1(z_0)]$ 的情况, 因此 $f(z) + g(z)$ 具有 **不少于 n 阶** 的零点.

(3) 代入得

$$\frac{f(z)}{g(z)} = (z - z_0)^{m-n} \frac{f_1(z)}{g_1(z)},$$

显然 $\frac{f_1(z_0)}{g_1(z_0)} \neq 0$, 故 $\frac{f(z)}{g(z)}$ 具有 $(m - n)$ 阶零点. 特别地, 当 $m = n$ 时, $z = z_0$ 为其 **可去奇点**.

10. (1) 由题意,

$$\begin{aligned} \frac{1}{z^2(1-z)} &= \frac{1}{z} \left(\frac{1}{z} + \frac{1}{1-z} \right) = \frac{1}{z^2} + \frac{1}{z} + \frac{1}{1-z} \\ &= z^{-2} + z^{-1} + \sum_{n=0}^{\infty} z^n = \sum_{n=-2}^{\infty} z^n. \end{aligned}$$

(2) 将 $e^{1/z}$ 在 $z = 0$ 附近展开成 Laurent 级数, 取曲线 C 为 $|z| = 1$, 则令 $u = \frac{1}{z}$, 对于 u 的积分路径同样为 $|u| = 1$. 需要注意的是, 若记 $z = e^{i\theta}$, 则 $u = e^{-i\theta}$, 而 θ 仍从 0 沿逆时针到 2π , 但 **u 的积分路径反向**, 故

$$a_n = \frac{1}{2\pi i} \int_C \frac{e^{1/z}}{z^{n+1}} dz \stackrel{u=1/z}{=} \frac{1}{2\pi i} \int_{-C} u^{n+1} e^u \cdot \frac{-du}{u^2} = \frac{1}{2\pi i} \int_C u^{n-1} e^u du.$$

i. 当 $n \geq 1$ 时, $u^{n-1} e^u$ 在 $|u| \leq 1$ 内解析, 由 Cauchy 积分定理, $a_n = 0$.

ii. 当 $n \geq 0$ 时, 记 $f(z) = e^z$, 则

$$a_n = \frac{1}{2\pi i} \int_C \frac{e^u}{u^{1-n}} du = \frac{1}{(-n)!} f^{(-n)}(0) = \frac{1}{(-n)!}.$$

综上, 该函数的 Laurent 级数为

$$z^2 e^{1/z} = z^2 \sum_{n=-\infty}^{\infty} a_n z^n = z^2 \sum_{n=-\infty}^0 \frac{z^n}{(-n)!} = \sum_{n=-\infty}^0 \frac{z^{n+2}}{(-n)!} = \sum_{n=-\infty}^2 \frac{z^n}{(2-n)!}.$$

另解 按提示, 考虑 e^u 在 $u = 0$ 附近的 Taylor 展开: $e^u = \sum_{n=0}^{\infty} \frac{u^n}{n!}$. 令 $u = \frac{1}{z}$, 则该函数的 Laurent 级数为

$$z^2 e^{1/z} = z^2 \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{1}{z} \right)^n \stackrel{m=-n+2}{=} z^2 \sum_{m=-\infty}^2 \frac{z^{m-2}}{(2-m)!} = \sum_{n=-\infty}^2 \frac{z^n}{(2-n)!}.$$

11. 首先将该函数展开成 $\frac{1}{a-b} \left(\frac{1}{z-a} - \frac{1}{z-b} \right)$ 的形式.

(1) $0 \leq |z| < |a|$ 时,

$$\frac{1}{z-a} = -\frac{1}{a} \sum_{n=0}^{\infty} \left(\frac{z}{a}\right)^n, \quad \frac{1}{z-b} = -\frac{1}{b} \sum_{n=0}^{\infty} \left(\frac{z}{b}\right)^n,$$

因此

$$\frac{1}{(z-a)(z-b)} = \frac{1}{a-b} \left(-\sum_{n=0}^{\infty} \frac{z^n}{a^{n+1}} + \sum_{n=0}^{\infty} \frac{z^n}{b^{n+1}} \right) = \frac{1}{b-a} \sum_{n=0}^{\infty} \left(\frac{1}{a^{n+1}} - \frac{1}{b^{n+1}} \right) z^n.$$

(2) $|a| < |z| < |b|$ 时,

$$\frac{1}{z-a} = \frac{1}{z} \sum_{n=0}^{\infty} \left(\frac{a}{z}\right)^n, \quad \frac{1}{z-b} = -\frac{1}{b} \sum_{n=0}^{\infty} \left(\frac{z}{b}\right)^n,$$

因此

$$\frac{1}{(z-a)(z-b)} = \frac{1}{a-b} \left(\sum_{n=0}^{\infty} \frac{a^n}{z^{n+1}} + \sum_{n=0}^{\infty} \frac{z^n}{b^{n+1}} \right) = \frac{1}{a-b} \left(\sum_{n=-\infty}^{-1} \frac{z^n}{a^{n+1}} + \sum_{n=0}^{\infty} \frac{z^n}{b^{n+1}} \right)$$

(3) 当 $|b| < |z| < \infty$ 时,

$$\frac{1}{z-a} = \frac{1}{z} \sum_{n=0}^{\infty} \left(\frac{a}{z}\right)^n, \quad \frac{1}{z-b} = \frac{1}{z} \sum_{n=0}^{\infty} \left(\frac{b}{z}\right)^n,$$

因此

$$\frac{1}{(z-a)(z-b)} = \frac{1}{a-b} \left(\sum_{n=0}^{\infty} \frac{a^n}{z^{n+1}} - \sum_{n=0}^{\infty} \frac{b^n}{z^{n+1}} \right) = \frac{1}{b-a} \sum_{n=-\infty}^{-1} \left(\frac{1}{a^{n+1}} - \frac{1}{b^{n+1}} \right) z^n.$$

(4) $0 < |z-a| < |b-a|$ 时,

$$\frac{1}{z-b} = \frac{1}{(z-a) - (b-a)} = -\frac{1}{b-a} \sum_{n=0}^{\infty} \left(\frac{z-a}{b-a}\right)^n,$$

因此

$$\frac{1}{(z-a)(z-b)} = \frac{1}{a-b} \left[\frac{1}{z-a} + \sum_{n=0}^{\infty} \frac{(z-a)^n}{(b-a)^{n+1}} \right] = -\sum_{n=-1}^{\infty} \frac{(z-a)^n}{(b-a)^{n+2}}.$$

(5) $|b-a| < |z-a| < \infty$ 时,

$$\frac{1}{z-b} = \frac{1}{(z-a) - (b-a)} = \frac{1}{z-a} \sum_{n=0}^{\infty} \left(\frac{b-a}{z-a}\right)^n,$$

因此

$$\begin{aligned} \frac{1}{(z-a)(z-b)} &= \frac{1}{a-b} \left[\frac{1}{z-a} - \sum_{n=0}^{\infty} \frac{(b-a)^n}{(z-a)^{n+1}} \right] = \sum_{n=1}^{\infty} \frac{(b-a)^{n-1}}{(z-a)^{n+1}} \\ &= \sum_{n=-\infty}^{-2} \frac{(z-a)^n}{(b-a)^{n+2}}. \end{aligned}$$

(6) $0 < |z - b| < |a - b|$ 时,

$$\frac{1}{z - a} = \frac{1}{(z - b) - (a - b)} = \frac{1}{b - a} \sum_{n=0}^{\infty} \left(\frac{z - b}{a - b} \right)^n,$$

因此

$$\frac{1}{(z - a)(z - b)} = \frac{1}{a - b} \left[- \sum_{n=0}^{\infty} \frac{(z - b)^n}{(a - b)^{n+1}} - \frac{1}{z - b} \right] = - \sum_{n=-1}^{\infty} \frac{(z - b)^n}{(a - b)^{n+2}}.$$

(7) $|a - b| < |z - b| < \infty$ 时,

$$\frac{1}{z - a} = \frac{1}{(z - b) - (a - b)} = \frac{1}{z - b} \sum_{n=0}^{\infty} \left(\frac{a - b}{z - b} \right)^n,$$

因此

$$\begin{aligned} \frac{1}{(z - a)(z - b)} &= \frac{1}{a - b} \left[\sum_{n=0}^{\infty} \frac{(a - b)^n}{(z - b)^{n+1}} - \frac{1}{z - b} \right] = \sum_{n=1}^{\infty} \frac{(a - b)^{n-1}}{(z - b)^{n+1}} \\ &= \sum_{n=-\infty}^{-2} \frac{(z - b)^n}{(a - b)^{n+2}}. \end{aligned}$$

12. (1) $f(z)$ 在 $a = a_i$ ($i = 1, 2, 3$) 附近的 Laurent 展开式为

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - a_i)^n = \sum_{n=1}^{\infty} \frac{a_{-n}}{(z - a_i)^n} + \sum_{n=0}^{\infty} a_n (z - a_i)^n.$$

当 a_i 为本性奇点时, $a_{-n} \neq 0$ ($\forall n \in \mathbb{N}^*$); 当 a_i 为 m 级极点时, $a_{-m} \neq 0$, 且当 $n > m$ 时, $a_{-n} \equiv 0$; 当 a_i 为可去奇点时, $a_{-n} \equiv 0$ ($\forall n \in \mathbb{N}^*$).

记在 $a = a_i$ 附近展开的 Laurent 级数的收敛半径为 R_i , 那么当 $a = a_1$ 时, $R_1 = |a_2 - a_1|$, 收敛域为 $0 < |z - a_1| < |a_2 - a_1|$; 当 $a = a_2$ 时, $R_2 = \min\{|a_1 - a_2|, |a_3 - a_2|\} = |a_2 - a_1|$, 收敛域为 $0 < |z - a_2| < |a_2 - a_1|$; 当 $a = a_3$ 时, $R_3 = |a_3 - a_2|$, 收敛域为 $0 < |z - a_3| < |a_3 - a_2|$.

(2) 当 a 为平面上其他点时, 可在 a 点作 Taylor 展开:

$$f(z) = \sum_{n=0}^{\infty} b_n (z - a)^n, \quad b_n = \left. \frac{d^n f(z)}{dz^n} \right|_{z=a}.$$

收敛半径 $R = \min_{1 \leq i \leq 3} |a_i - a|$, 则收敛域为 $|z - a| < R$.

13. (1) 变形得 $\frac{e^z}{z^2 + 4} = \frac{e^z}{(z + 2i)(z - 2i)}$, 而 $e^{\pm 2i} \neq 0$, 因此 $\pm 2i$ 均为 1 级极点.

(2) 奇点满足 $\cos z = 0$, 即 $z_k = \left(k + \frac{1}{2}\right)\pi \in \mathbb{R}$ ($k \in \mathbb{Z}$). 记 $f(z) = \frac{1}{\cos z}$, 则 z_k 为 $\frac{1}{f(z)} = \cos z$ 的 1 级零点, 因此 z_k 为 $\frac{1}{\cos z}$ 的 1 级极点.

(3) 奇点为 $z_0 = 1$, 将函数在 z_0 附近作 Laurent 展开:

$$\sin \frac{1}{1-z} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \frac{1}{(1-z)^{2n+1}},$$

可见展开式的主要部分有无穷多项, 因此 $z_0 = 1$ 为其本性奇点.

(4) 奇点为 $z_k = 2k\pi i$ ($k \in \mathbb{Z}$), 记 $f(z) = \frac{1}{1-e^z}$, 则

$$\left. \frac{d}{dz} \left(\frac{1}{f(z)} \right) \right|_{z=z_k} = \left. \frac{d}{dz} (1-e^z) \right|_{z=z_k} = -1,$$

因此 $z = z_k$ 为 $\frac{1}{f(z)}$ 的 1 级零点, 故 $z = z_k$ 为 $f(z)$ 的 1 级极点.

(5) 奇点为 $z_0 = 0$, 取 $z = \operatorname{Re}(z) = x$, 则

$$\lim_{z \rightarrow 0} e^{-z} \cos \frac{1}{z} = 1 \cdot \lim_{x \rightarrow 0} \cos \frac{1}{x},$$

而后者极限不存在, 故 $\lim_{z \rightarrow 0} e^{-z} \cos \frac{1}{z}$ 极限必然不存在, $z_0 = 0$ 为其本性奇点.

(6) 奇点为 $z_1 = 0, z_2 = 1$, 又因为

$$\lim_{z \rightarrow 0} \frac{e^z - 1}{z} = \lim_{z \rightarrow 0} \frac{1}{z} \sum_{k=1}^{\infty} \frac{z^k}{k!} = \lim_{z \rightarrow 0} \sum_{k=1}^{\infty} \frac{z^{k-1}}{k!} = 1,$$

因此在 $z_1 = 0$ 附近,

$$\lim_{z \rightarrow 0} \frac{z}{e^z - 1} \exp \left(\frac{1}{z-1} \right) = 1 \cdot e^{-1} = \frac{1}{e},$$

即 $z_1 = 1$ 为其可去奇点. 取 $z = \operatorname{Re}(z) = x$, 则

$$\begin{aligned} \lim_{x \rightarrow 1^+} \frac{z}{e^z - 1} \exp \left(\frac{1}{z-1} \right) &= \frac{1}{e-1} \lim_{x \rightarrow 1^+} \exp \left(\frac{1}{x-1} \right) = +\infty, \\ \lim_{x \rightarrow 1^-} \frac{z}{e^z - 1} \exp \left(\frac{1}{z-1} \right) &= \frac{1}{e-1} \lim_{x \rightarrow 1^-} \exp \left(\frac{1}{x-1} \right) = 0, \end{aligned}$$

因此 $\lim_{x \rightarrow 1} \frac{z}{e^z - 1} \exp \left(\frac{1}{z-1} \right)$ 极限不存在, 故 $z \rightarrow 1$ 时其极限必然不存在.

综上, $z_1 = 0, z_2 = 1$ 分别为其可去奇点与本性奇点.

(7) 奇点为 $z_1 = 0, z_2 = -1, z_3 = 3$, 注意到

$$\lim_{z \rightarrow 0} \frac{\sin z}{z} = \lim_{z \rightarrow 0} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} z^{2k} = 1,$$

因此在 $z_1 = 0$ 附近,

$$\frac{\sin z}{(z-3)^2 z^2 (z+1)^3} = \frac{1}{z} \left[\frac{1}{(z-3)^2 (z+1)^3} \frac{\sin z}{z} \right],$$

故 $z_1 = 0$ 为其 1 级极点. 由于 $\sin z_2, \sin z_3 \neq 0$, z_2, z_3 分别为其 3 级、2 级极点.

(8) $\sin z - \sin a = 2 \cos \frac{z+a}{2} \sin \frac{z-a}{2}$, 因此奇点为

$$z_k = (2k+1)\pi - a, \quad z_l = 2l\pi + a, \quad k, l \in \mathbb{Z}.$$

记 $f(z) = \frac{1}{\sin z - \sin a}$, 那么

$$\begin{aligned} \left. \frac{d}{dz} \left[\frac{1}{f(z)} \right] \right|_{z=z_k} &= \left. \frac{d}{dz} (\sin z - \sin a) \right|_{z=z_k} = \cos z_k = -\cos a, \\ \left. \frac{d}{dz} \left[\frac{1}{f(z)} \right] \right|_{z=z_l} &= \left. \frac{d}{dz} (\sin z - \sin a) \right|_{z=z_l} = \cos z_l = \cos a, \end{aligned}$$

因此若 $a = \left(m + \frac{1}{2}\right)\pi$, 则 z_k, z_l 均为其 2 级极点; 若 a 为其他取值, 则 z_k, z_l 均为其 1 级极点.

(9) 奇点为 $z_0 = 0$, 将 $(1 - \cos z)$ 在 z_0 附近作 Taylor 展开:

$$\frac{1 - \cos z}{z^n} = \frac{1}{z^n} \left[1 - \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} z^{2k} \right] = - \sum_{k=1}^{\infty} \frac{(-1)^k}{(2k)!} z^{2k-n},$$

因此若 $n < 2$, 则 $2k - n > 0$, 即 $\lim_{z \rightarrow 0} \frac{1 - \cos z}{z^n} = 0$; 若 $n = 2$, 则 $k = 1$ 时 $2k - n = 0$, 即 $\lim_{z \rightarrow 0} \frac{1 - \cos z}{z^2} = \frac{1}{2}$; 若 $n > 2$, 则存在 z 的负幂次项, 负幂次最高为 $-(2-n) = n-2$, 此时 z_0 为其 $(n-2)$ 级极点.

综上, 若 $n \leq 2$, 则 $z_0 = 0$ 为其可去奇点; 若 $n \geq 3$, 则 $z_0 = 0$ 为其 $(n-2)$ 级极点.

14. (1) $\lim_{z \rightarrow \infty} \frac{z^2}{2 + z^2} = 1$, 因此 ∞ 为其可去奇点.

(2) 取 $z = \operatorname{Re}(z) = x$, 则

$$\lim_{x \rightarrow +\infty} \frac{x^2 + 4}{e^x} = 0, \quad \lim_{x \rightarrow -\infty} \frac{x^2 + 4}{e^x} = +\infty,$$

因此 $\lim_{x \rightarrow \infty} \frac{z^2 + 4}{e^z}$ 极限不存在, 故 $\lim_{z \rightarrow 0} \frac{z^2 + 4}{e^z}$ 必然不存在, ∞ 为其本性奇点.

(3) 令 $u = \frac{1}{z}$, 那么

$$\lim_{z \rightarrow \infty} \exp(-z^{-2}) = \lim_{u \rightarrow 0} e^{-u^2} = 1,$$

因此 ∞ 为其可去奇点.

(4) 将其在 $0 < |z| < \infty$ 范围内作 Laurent 展开, 得

$$\frac{1 - \cos z}{z^n} = \frac{1}{z^n} \left[1 - \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} z^{2k} \right] = - \sum_{k=1}^{\infty} \frac{(-1)^k}{(2k)!} z^{2k-n},$$

可见无论 n 取何值, 当 $k \geq \max \left\{ \left[\frac{n}{2} + 1 \right], 0 \right\}$ 时, 该函数的展开式总有无穷多个正幂次项, 因此 ∞ 为其本性奇点.

(5) 令 $u = \frac{1}{z}$, 那么 $\frac{z^5}{z^2+8} = \frac{1}{(1+8u^2)u^3}$; 而 $u_0 = 0$ 为 $\frac{1}{(1+8u^2)u^3}$ 的 3 级极点, 故 ∞ 为 $\frac{z^5}{z^2+8}$ 的 3 级极点.

(6) 令 $u = \frac{1}{z}$, 那么 $\sec \frac{1}{z} = \frac{1}{\cos u}$. 由第 13(2) 题结论, $u_0 = 0$ 为其 1 级极点, 故 ∞ 为 $\sec \frac{1}{z}$ 的 1 级极点.

(7) 令 $u = \frac{1}{z}$, 那么 $\sin \frac{1}{z} = \sin u$, 而 $\lim_{u \rightarrow 0} \sin u = 0$, 故 u_0 为 $\sin u$ 的可去奇点, 因此 ∞ 为 $\sin \frac{1}{z}$ 的可去奇点.

(8) a

(9) 令 $z = \operatorname{Re}(z) = x$, 则

$$\begin{aligned}\lim_{x \rightarrow +\infty} e^{-z} \cos \frac{1}{z} &= \lim_{x \rightarrow +\infty} e^{-x} \cos \frac{1}{x} = 0 \cdot 1 = 0, \\ \lim_{x \rightarrow -\infty} e^{-z} \cos \frac{1}{z} &= \lim_{x \rightarrow -\infty} e^{-x} \cos \frac{1}{x} = +\infty \cdot 1 = +\infty,\end{aligned}$$

因此 $\lim_{z \rightarrow \infty} e^{-z} \cos \frac{1}{z}$ 不存在, ∞ 为其本性奇点.

15. 由题意, $\exists f_1(z), g_1(z)$ 在 a 点解析且 $f_1(a), g_1(a) \neq 0$, 并且

$$f(z) = \frac{f_1(z)}{(z-a)^m}, \quad g(z) = \frac{g_1(z)}{(z-a)^n}.$$

(1) 若 $m > n$, 则

$$f(z) \pm g(z) = \frac{f_1(z) \pm (z-a)^{m-n} g_1(z)}{(z-a)^m},$$

显然 $[f_1(z) \pm (z-a)^{m-n} g_1(z)]$ 在 $z = 0$ 处非零, 故 $z = a$ 为其 $(m-n)$ 级极点; $m < n$ 时同理.

而当 $m = n$ 时, $f_1(z) \pm g_1(z)$ 可能被 $(z-a)$ 整除, a 为 $f(z) \pm g(z)$ 的至多 m 级极点.

综上, 当 $m \neq n$ 时, a 为 $f(z) \pm g(z)$ 的 $(\max\{m, n\} - \max\{m, n\})$ 级极点; 当 $m = n$ 时, a 为 $f(z) \pm g(z)$ 的至多 m 级极点或可去极点.

(2) $f(z)g(z) = \frac{f_1(z)g_1(z)}{(z-a)^{m+n}}$, 显然 $f_1(a)g_1(a) \neq 0$ 且 $f_1(z)g_1(z)$ 在 $z = a$ 附近解析, 故 $z = a$ 为其 $(m+n)$ 级极点.

(3) 当 $m > n$ 时,

$$\frac{f(z)}{g(z)} = \frac{f_1(z)}{g_1(z)} \frac{1}{(z-a)^{m-n}},$$

显然 $\frac{f_1(z)}{g_1(z)}$ 在 $z = a$ 处非零且在其附近解析, 故 a 为 $\frac{f(z)}{g(z)}$ 的 $(m-n)$ 级极点.

当 $m \leq n$ 时, $\frac{f(z)}{g(z)} = \frac{f_1(z)}{g_1(z)}(z-a)^{n-m}$, 即

$$\lim_{z \rightarrow a} \frac{f(z)}{g(z)} = \frac{f_1(a)}{g_1(a)} \lim_{z \rightarrow a} (z-a)^{n-m} = \begin{cases} \frac{f_1(a)}{g_1(a)}, & m = n, \\ 0, & m < n, \end{cases}$$

此时 $z = a$ 为 $\frac{f(z)}{g(z)}$ 的可去奇点.

16. 由题意, 设

$$f(z) = \frac{\varphi(z)}{(z-a)^m}, \quad \varphi(a) \neq 0, \quad m \in \mathbb{N}^*,$$
$$g(z) = \sum_{n=-\infty}^{\infty} a_n(z-a)^n, \quad a_n \neq 0 \quad (n \leq -1).$$

(1) 可知

$$f(z)g(z) = \varphi(z) \sum_{n=-\infty}^{\infty} a_n(z-a)^{n-m}$$

5 留数及其应用

1. (1) $z_0 = i$ 为其 1 级极点, 那么

$$\operatorname{Res}\left[\frac{\cos z}{z-i}, z_0\right] = \frac{\cos i}{1} = \cosh 1.$$

(2) 奇点满足 $z^{2n} = -1$, 故 $z_k = \exp\left[i\frac{(2k+1)\pi}{2n}\right]$ ($k = 0, 1, \dots, 2n-1$) 为其 1 级极点,

$$\operatorname{Res}\left[\frac{z^{2n}}{1+z^{2n}}, z_k\right] = \frac{z_k^{2n}}{2n \cdot z_k^{2n-1}} = \frac{z_k}{2n} = \frac{1}{2n} \exp\left[i\frac{(2k+1)\pi}{2n}\right], \quad k = 0, 1, \dots, 2n-1.$$

(3) $z_k = 2k\pi i$ ($k \in \mathbb{Z}$) 为其 1 级极点, 那么

$$\operatorname{Res}\left[\frac{1}{e^z - 1}, z_k\right] = \frac{1}{e^{z_k}} = 1.$$

(4) $z_0 = 0$ 为其 4 级极点, 那么

$$\operatorname{Res}\left[\frac{1 - e^{2z}}{z^4}, z_0\right] = \frac{1}{3!} \lim_{z \rightarrow 0} \frac{d^3}{dz^3} \left[z^4 \cdot \frac{1 - e^{2z}}{z^4} \right] = -\frac{4}{3}.$$

(5) $\frac{1}{(1+z^2)^3} = \frac{1}{(z-i)^3(z+i)^3}$, 因此 $z_1 = -i, z_2 = i$ 均为其 3 级极点, 那么

$$\begin{aligned} \operatorname{Res}\left[\frac{1}{(1+z^2)^3}, z_1\right] &= \frac{1}{2!} \lim_{z \rightarrow -i} \frac{d^2}{dz^2} \left[(z+i)^3 \cdot \frac{1}{(1+z^2)^3} \right] = \frac{1}{2} \cdot \frac{12}{(-2i)^5} = \frac{3}{16}i, \\ \operatorname{Res}\left[\frac{1}{(1+z^2)^3}, z_2\right] &= \frac{1}{2!} \lim_{z \rightarrow i} \frac{d^2}{dz^2} \left[(z-i)^3 \cdot \frac{1}{(1+z^2)^3} \right] = \frac{1}{2} \cdot \frac{12}{(2i)^5} = -\frac{3}{16}i. \end{aligned}$$

(6) $z_0 = 1$ 为其 n 级极点, 那么

$$\operatorname{Res}\left[\frac{z^{2n}}{(z-1)^n}, z_0\right] = \frac{1}{(n-1)!} \lim_{z \rightarrow 1} \frac{d^{n-1}}{dz^{n-1}} \left[(z-1)^n \cdot \frac{z^{2n}}{(z-1)^n} \right] = \frac{2n!}{(n-1)!(n+1)!}.$$

(7) z_1, z_2 分别为其 m, n 级极点, 那么

$$\begin{aligned} \operatorname{Res}\left[\frac{1}{(z-z_1)^m(z-z_2)^n}, z_1\right] &= \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} \left[(z-z_1)^m \cdot \frac{1}{(z-z_1)^m(z-z_2)^n} \right] \\ &= \frac{(n+m-2)!}{(n-1)!(m-1)!} \frac{(-1)^{m-1}}{(z_1-z_2)^{n+m-1}}, \\ \operatorname{Res}\left[\frac{1}{(z-z_1)^m(z-z_2)^n}, z_2\right] &= \frac{1}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} \left[(z-z_2)^n \cdot \frac{1}{(z-z_1)^m(z-z_2)^n} \right] \\ &= \frac{(n+m-2)!}{(n-1)!(m-1)!} \frac{(-1)^{n-1}}{(z_2-z_1)^{n+m-1}}. \end{aligned}$$

(8) 注意到

$$\frac{1}{z+1} + \dots + \frac{1}{(z+1)^n} = \frac{1}{z+1} \frac{1 - \left(\frac{1}{z+1}\right)^n}{1 - \frac{1}{z+1}} = \frac{1}{z} \left[1 - \frac{1}{(z+1)^n} \right],$$

因此原函数 $f(z)$ 可写为

$$f(z) = \frac{1}{z} \left[\frac{1}{z+1} + \cdots + \frac{1}{(z+1)^n} \right] = \frac{1}{z^2} \left[1 - \frac{1}{(z+1)^n} \right],$$

$z_1 = 0, z_2 = -1$ 分别为其 2 级、 n 级极点, 那么

$$\begin{aligned} \operatorname{Res}[f(z), z_1] &= \frac{1}{1!} \lim_{z \rightarrow 0} \frac{d}{dz} [z^2 f(z)] = \frac{n}{(0+1)^n} = n, \\ \operatorname{Res}[f(z), z_2] &= \frac{1}{(n-1)!} \lim_{z \rightarrow -1} \frac{d^{n-1}}{dz^{n-1}} [(z+1)^n f(z)] \\ &\stackrel{\text{Leibniz 公式}}{=} \frac{1}{(n-1)!} \lim_{z \rightarrow -1} \left\{ [(z+1)^n - 1] \cdot \left(\frac{1}{z^2} \right)^{(n-1)} + 0 \right\} \\ &= \frac{1}{(n-1)!} \cdot (-1) \cdot (-1)^{n-1} \frac{n!}{(-1)^{n+1}} = -n. \end{aligned}$$

2. (1) 记 $\zeta = \frac{1}{z}$, 则 $\varphi(\zeta) = f(z)$, 其收敛域为 $0 < |\zeta| < \frac{1}{R}$, 那么 $\varphi(\zeta)$ 在收敛域内的 Laurent 展开式为

$$\varphi(\zeta) = \sum_{n=-\infty}^{\infty} a_n \zeta^n, \quad a_n = \frac{1}{2\pi i} \int_{C^+} \frac{\varphi(\zeta)}{\zeta^{n+1}} d\zeta,$$

其中 C^+ 为区域内任一包含原点且取正向的闭路, 因此

$$f(z) = \varphi\left(\frac{1}{z}\right) = \sum_{n=-\infty}^{\infty} a_n \left(\frac{1}{z}\right)^n = \sum_{n=-\infty}^{\infty} b_n z^n,$$

注意到换元时路径方向反向⁵, 那么

$$b_n = a_{-n} = \frac{1}{2\pi i} \int_{C^+} \frac{\varphi(\zeta)}{\zeta^{-n+1}} d\zeta \stackrel{\zeta=1/z}{=} -\frac{1}{2\pi i} \int_{C^-} \frac{f(z)}{z^{n+1}} dz = -\frac{1}{2\pi i} \int_{C^-} \frac{f(z)}{z^{n+1}} dz,$$

因此当 b_{-1} 恰为 $f(z)$ 在无穷远处的留数的相反数, 即

$$\operatorname{Res}[f(z), \infty] = -b_{-1} = \frac{1}{2\pi i} \int_{C^-} f(z) dz.$$

- (2) 取一条包含 $a_1, a_2, \dots, a_n$ 的闭路 C , 结合留数定理,

$$\begin{aligned} \sum_{k=1}^n \operatorname{Res}[f(z), a_k] + \operatorname{Res}[f(z), \infty] &= \frac{1}{2\pi i} \int_{C^+} f(z) dz + \frac{1}{2\pi i} \int_{C^-} f(z) dz \\ &= \frac{1}{2\pi i} \int_{C^+ + C^-} f(z) dz = 0. \end{aligned}$$

- (3) 对于函数 $\frac{\cos z}{z-i}, \frac{1}{(1+z^2)^3}$, 由上一问结论,

$$\begin{aligned} \operatorname{Res}\left[\frac{\cos z}{z-i}, \infty\right] &= -\operatorname{Res}\left[\frac{\cos z}{z-i}, i\right] = -\cosh 1, \\ \operatorname{Res}\left[\frac{1}{(1+z^2)^3}, \infty\right] &= -\operatorname{Res}\left[\frac{1}{(1+z^2)^3}, -i\right] - \operatorname{Res}\left[\frac{1}{(1+z^2)^3}, i\right] = 0. \end{aligned}$$

⁵ 设想 $u = e^{i\theta}$, 其中 θ 从 0 到 2π ; 而 $z = \frac{1}{u} = e^{-i\theta} = e^{i\varphi}$, 其中 φ 从 0 到 -2π , 可见路径反向

对于函数 $\sin \frac{1}{z}, e^{1/z}$, 将其在 ∞ 附近作 Laurent 展开, 得、

$$\sin \frac{1}{z} = \sum_{k=0}^{\infty} (-1)^k \frac{1}{(2k+1)!} \frac{1}{z^{2k+1}}, \quad e^{1/z} = \sum_{k=0}^{\infty} \frac{1}{k!} \frac{1}{z^k},$$

由第一小问结论, 两者的 b_{-1} 均为 1, 因此

$$\operatorname{Res}\left[\sin \frac{1}{z}, \infty\right] = -b_{-1} = -1, \quad \operatorname{Res}\left[e^{1/z}, \infty\right] = -b_{-1} = -1.$$

(4) 注意到

$$\cos \frac{1}{z-1} = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k}}{(2k)!},$$

即其在 ∞ 附近的 Laurent 展开式中 $(z-1)^{-1}$ 的系数为 0, 故

$$\operatorname{Res}\left[\cos \frac{1}{z-1}, \infty\right] = -b_{-1} = 0.$$

由上一问结论及积分的可加性,

$$\begin{aligned} \operatorname{Res}\left[\sin \frac{z}{1-z}, 1\right] &= -\operatorname{Res}\left[\sin\left(\frac{1}{1-z} - 1\right), \infty\right] \\ &= \cos 1 \operatorname{Res}\left[\sin \frac{1}{z-1}, \infty\right] + \sin 1 \operatorname{Res}\left[\cos \frac{1}{z-1}, \infty\right] \\ &= \cos 1 \cdot (-1) + \sin 1 \cdot 0 = -\cos 1. \end{aligned}$$

3. (1) 圆 $C: (x-1)^2 + (y-1)^2 = 2$, 因此 $z_1 = 1, z_2 = i$ 均在 C 所包围的区域内⁶, 分别为函数的 2 级、1 级极点, 而 $z_3 = -i$ 并非奇点. 由留数定理,

$$\begin{aligned} \int_C \frac{dz}{(z-1)^2(z^2+1)} &= 2\pi i \sum_{k=1}^2 \operatorname{Res}\left[\frac{1}{(z-1)^2(z^2+1)}, z_k\right] \\ &= 2\pi i \cdot \lim_{z \rightarrow 1} \frac{d}{dz} \left[(z-1)^2 \cdot \frac{1}{(z-1)^2(z^2+1)} \right] + 2\pi i \cdot \frac{1/(i-1)^2(2i)}{1} \\ &= 2\pi i \cdot \left(-\frac{1}{2} + \frac{1}{4}\right) = -\frac{\pi}{2}i. \end{aligned}$$

- (2) 圆 $C: (x-1)^2 + y^2 = 1$, 因此函数在全平面的奇点 $z_k = \exp\left[i\frac{(2k+1)\pi}{4}\right]$ ($k = 0, 1, 2, 3$) 中, z_0, z_3 位于 C 所包围的区域内, 且为 1 级极点. 由留数定理,

$$\begin{aligned} \int_C \frac{dz}{z^4+1} &= 2\pi i \operatorname{Res}\left[\frac{1}{z^4+1}, z_0\right] + 2\pi i \operatorname{Res}\left[\frac{1}{z^4+1}, z_3\right] \\ &= 2\pi i \left(\prod_{i=1}^3 \frac{1}{z_0 - z_i} + \prod_{j=0}^2 \frac{1}{z_3 - z_j} \right) \\ &= 2\pi i \cdot \left[\frac{1}{\sqrt{2} \cdot \sqrt{2}(1+i) \cdot \sqrt{2}i} + \frac{1}{-\sqrt{2}i \cdot \sqrt{2}(1-i)\sqrt{2}} \right] = -\frac{\pi}{\sqrt{2}}i. \end{aligned}$$

⁶ $z_2 = i$ 对应 $x = 0, y = 1$, 后 $z_3 = -i$ 同理

(3) 此函数在全平面的奇点有 $z_k = \exp \left[i \frac{(2k+1)\pi}{3} \right]$ ($k = 0, 1, 2$), $z_3 = 1$, 且 $|z_n| = 1$ ($n = 0, 1, 2, 3$).

i. $r < 1$ 时, 此函数在 C 包围的域内解析, 由 Cauchy 积分定理,

$$\int_C \frac{dz}{(z^2-1)(z^3+1)} = 0.$$

ii. $r > 1$ 时, z_0, z_2, z_3 均为其 1 级极点, z_1 为其 2 级极点, 那么⁷

$$\begin{aligned} \int_C \frac{dz}{(z^2-1)(z^3+1)} &= 2\pi i \left[\sum_{i \neq 1} \frac{1}{(z_i+1)^2} \prod_{j \neq 1, i} \frac{1}{z_i - z_j} + \lim_{z \rightarrow -1} \frac{d}{dz} \left(\prod_{i \neq 1} \frac{1}{z - z_i} \right) \right] \\ &= 2\pi i \left[\frac{e^{i\frac{\pi}{2}}}{3\sqrt{3}} + \frac{e^{-i\frac{\pi}{2}}}{3\sqrt{3}} + \frac{1}{4} - \frac{1}{6} \left(\frac{e^{-i\frac{\pi}{6}}}{\sqrt{3}} + \frac{e^{i\frac{\pi}{6}}}{\sqrt{3}} + \frac{1}{2} \right) \right] = 0. \end{aligned}$$

综上,

$$\int_C \frac{dz}{(z^2-1)(z^3+1)} = 0.$$

另解 $r > 1$ 时, 记正向闭路 $C_R: R < |z| < r$, 其中 $1 < R < r$, 那么由复闭路的 Cauchy 积分定理,

$$\int_{C_R} \frac{dz}{(z^2-1)(z^3+1)} = \int_C \frac{dz}{(z^2-1)(z^3+1)}.$$

将被积函数在 $1 < R < |z| < r$ 内作 Laurent 展开:

$$\frac{1}{(z^2-1)(z^3+1)} = \frac{1}{z^2} \sum_{m=0}^{\infty} \left(\frac{1}{z^2} \right)^m \cdot \frac{1}{z^3} \sum_{n=0}^{\infty} \left(-\frac{1}{z^3} \right)^n = \sum_{m,n=0}^{\infty} \frac{(-1)^n}{z^{2m+3n+5}}.$$

可见 $2m+3n+5 \geq 5 > 1$, 因此 $a_{-1} = 0$, 即

$$\int_C \frac{dz}{(z^2-1)(z^3+1)} = \int_{C_R} \frac{dz}{(z^2-1)(z^3+1)} = 2\pi i a_{-1} = 0.$$

(4) $z_k = k$ ($k = 1, 2, 3$) 均为其在 C 所围成的闭域内的 1 级极点, 因此

$$\int_C \frac{dz}{(z-1)(z-2)(z-3)} = 2\pi i \left(\frac{1}{2} - 1 + \frac{1}{2} \right) = 0.$$

(5) $z_1 = 1, z_2 = -1$ 均为其在 C 所包围的闭域内的 2 级极点, 而 $z_3 = 3$ 不在此内, 由留数定理,

$$\begin{aligned} \int_C \frac{dz}{(z^2-1)(z-3)^2} &= 2\pi i \left\{ \lim_{z \rightarrow 1} \frac{d}{dz} \left[\frac{1}{(z+1)^2(z-3)^2} \right] + \lim_{z \rightarrow -1} \frac{d}{dz} \left[\frac{1}{(z-1)^2(z-3)^2} \right] \right\} \\ &= 2\pi i \cdot \left(0 + \frac{3}{128} \right) = \frac{3\pi}{64} i. \end{aligned}$$

⁷技巧: 在各项的复数运算中, 加减用一般式, 乘除用指数式

4. (1) 记 $z = e^{i\theta}$ ($0 \leq \theta < 2\pi$), 则 $dz = ie^{i\theta} d\theta = iz d\theta$. 记正向闭路 $C: |z| = 1$, 那么

$$\begin{aligned}\int_0^{2\pi} \frac{d\theta}{a + \cos \theta} &= \int_C \frac{1}{a + \frac{1}{2}\left(z + \frac{1}{z}\right)} \frac{dz}{iz} = \frac{2}{i} \int_C \frac{dz}{z^2 + 2az + 1} \\ &= \frac{2}{i} \int_C \frac{dz}{(z + a - \sqrt{a^2 - 1})(z + a + \sqrt{a^2 - 1})^2}.\end{aligned}$$

由于 $a > 1$, 故

$$\begin{aligned}|z_1| &= \left| -a - \sqrt{a^2 - 1} \right| = a + \sqrt{a^2 - 1} > 1 + 0 = 1, \\ |z_2| &= \left| -a + \sqrt{a^2 - 1} \right| = \frac{1}{a + \sqrt{a^2 - 1}} < \frac{1}{1} = 1,\end{aligned}$$

因此 $z_2 = -a + \sqrt{a^2 - 1}$ 为 C 所包围的区域内的 1 级极点, 故

$$\begin{aligned}\int_0^{2\pi} \frac{d\theta}{a + \cos \theta} &= \frac{2}{i} \int_C \frac{dz}{z^2 + 2az + 1} = 2\pi i \cdot \frac{2}{i} \cdot \frac{1}{z^2 + a + \sqrt{a^2 - 1}} \\ &= \frac{2\pi}{\sqrt{a^2 - 1}}.\end{aligned}$$

(2) 记 $z = e^{i\theta}$ ($0 \leq \theta < 2\pi$), 正向闭路 $C: |z| = 1$, 那么

$$\begin{aligned}\int_0^{2\pi} \frac{r - \cos \theta}{1 - 2r \cos \theta + r^2} d\theta &= \frac{1}{2} \int_C \frac{2rz - (z^2 + 1)}{(1 + r^2)z - r(z^2 + 1)} \frac{dz}{iz} \\ &= \frac{1}{2i} \int_C \frac{z^2 - 2rz + 1}{z(z - r)(rz - 1)} dz.\end{aligned}$$

i. 当 $r > 1$ 时, $z_0 = 0, z_1 = \frac{1}{r}$ 为被积函数在 C 所包围的区域内的 1 级极点, 那么

$$\int_0^{2\pi} \frac{r - \cos \theta}{1 - 2r \cos \theta + r^2} d\theta = 2\pi i \cdot \frac{1}{2i} \left(\frac{1}{r} + \frac{1}{r} \right) = \frac{2\pi}{r}.$$

ii. 当 $r < 1$ 时, $z_0 = 0, z_2 = r$ 为被积函数在 C 所包围的区域内的 1 级极点, 那么

$$\int_0^{2\pi} \frac{r - \cos \theta}{1 - 2r \cos \theta + r^2} d\theta = 2\pi i \cdot \frac{1}{2i} \left(\frac{1}{r} - \frac{1}{r} \right) = 0.$$

iii. 当 $r = 1$ 时, 该积分发散:

$$\int_0^{2\pi} \frac{r - \cos \theta}{1 - 2r \cos \theta + r^2} d\theta = \frac{1}{2i} \int_C \frac{dz}{z} = \frac{1}{2i} \cdot 2\pi i \cdot 1 = \pi.$$

综上,

$$\int_0^{2\pi} \frac{r - \cos \theta}{1 - 2r \cos \theta + r^2} d\theta = \begin{cases} 0, & r < 1, \\ \pi, & r = 1, \\ \frac{2\pi}{r}, & r > 1 \end{cases} = \frac{\pi}{r} [1 + \operatorname{sgn}(r - 1)].$$

(3) 令 $z = e^{i(2\theta)}$ ($0 \leq \theta < \pi$), 则 $dz = 2ie^{i(2\theta)} d\theta = 2iz d\theta$, 且

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2} = \frac{1}{4} \left(z + \frac{1}{z} + 2 \right),$$

记正向曲线 $C: |z| = 1$, 因此

$$\begin{aligned} \int_0^{\pi/2} \frac{d\theta}{a^2 + \sin^2 \theta} &= \frac{1}{2} \int_0^\pi \frac{d\theta}{a^2 + 1 - \cos^2 \theta} = \frac{1}{2} \int_C \frac{1}{a^2 + 1 - \frac{1}{4} \left(z + \frac{1}{z} + 2 \right)} \frac{dz}{2iz} \\ &= \frac{1}{-i} \int_C \frac{dz}{z^2 - 2(2a^2 + 1)z + 1}. \end{aligned}$$

可见被积函数奇点为

$$z_1 = (2a^2 + 1) - 2a\sqrt{a^2 + 1} \in \mathbb{R}, \quad z_2 = (2a^2 + 1) + 2a\sqrt{a^2 + 1} \in \mathbb{R},$$

又因为

$$\begin{aligned} |z_1| &= z_1 = (2a^2 + 1) - 2a\sqrt{a^2 + 1} > 0 + 1 + 0 = 1, \\ |z_1| &= z_2 = (2a^2 + 1) - 2a\sqrt{a^2 + 1} = \frac{1}{(2a^2 + 1) + 2\sqrt{a^2 + 1}} < 1, \end{aligned}$$

因此 z_1 在 C 所包围的区域内, 为奇点, 而 z_2 并非奇点, 由留数定理,

$$\begin{aligned} \int_0^{\pi/2} \frac{d\theta}{a^2 + \sin^2 \theta} &= \frac{1}{-i} \int_C \frac{dz}{(z - z_1)(z - z_2)} = 2\pi i \cdot \frac{1}{-i(z_1 - z_2)} \\ &= -2\pi \cdot \frac{1}{-4a\sqrt{a^2 + 1}} = \frac{\pi}{2a\sqrt{1 + a^2}}. \end{aligned}$$

(4) 记 $\varphi = \theta + \pi$, 那么

$$\int_0^\pi \tan(\theta + ia) d\theta \stackrel{\varphi = \theta + \pi}{=} \int_\pi^{2\pi} \tan(\varphi - \pi + ia) d\varphi = \int_\pi^{2\pi} \tan(\varphi + ia) d\varphi.$$

记 $z = e^{i(\theta + ia)} = e^{-a} e^{i\theta}$ ($0 \leq \theta < 2\pi$), 正向曲线 $C: |z| = e^{-a}$, 那么

$$\begin{aligned} \int_0^\pi \tan(\theta + ia) d\theta &= \frac{1}{2} \int_0^{2\pi} \frac{\sin(\theta + ia)}{\cos(\theta + ia)} d\theta = \frac{1}{2} \int_C \frac{z - \frac{1}{z}}{i \left(z + \frac{1}{z} \right)} \frac{dz}{iz} \\ &= \frac{1}{2} \int_C \frac{1 - z^2}{z(1 + z^2)} dz \end{aligned}$$

i. $a > 0$ 时, 闭路 C 上 $|z| = e^{-a} < 1$, 因此被积函数在全平面的奇点 $z_1 = -i, z_2 = i$ 均位于 C 所包围的区域外, 而 $z_0 = 0$ 为其 1 级极点, 由留数定理,

$$\int_C \tan(\theta + ia) d\theta = \frac{1}{2} \cdot 2\pi i \cdot \frac{1 - 0}{1 + 0} = \pi i.$$

ii. $a < 0$ 时, 闭路 C 上 $|z| = e^{-a} > 1$, 因此在 C 所包围的区域内具有奇点 z_0, z_1, z_2 , 且 z_0, z_1, z_2 均为 1 级极点, 由留数定理,

$$\begin{aligned} \int_0^\pi \tan(\theta + ia) d\theta &= \frac{1}{2} \int_C \frac{1 - z^2}{z(1 + z^2)} dz = 2\pi i \cdot \frac{1}{2} \left[\frac{2}{i \cdot 2i} + \frac{2}{(-i) \cdot (-2i)} + \frac{1 - 0}{1 + 0} \right] \\ &= -\pi i. \end{aligned}$$

综上,

$$\int_0^\pi \tan(\theta + ia) d\theta = \begin{cases} -\pi i, & a < 0, \\ \pi i, & a > 0 \end{cases} = \pi i \operatorname{sgn}(a).$$

5. (1) 记 $f(z) = \frac{z^2}{(z^2 + a^2)^2}$, 则

$$\lim_{z \rightarrow \infty} z f(z) = \lim_{z \rightarrow \infty} \frac{z^3}{(z^2 + a^2)^2} = 0,$$

那么

$$\int_{-\infty}^{\infty} \frac{x^2}{(x^2 + a^2)^2} dx = 2\pi i \operatorname{Res}[f(z), ai] = 2\pi i \lim_{z \rightarrow ai} \frac{d}{dz} \left[\frac{z^2}{(z + ai)^2} \right] = \frac{\pi}{2a}.$$

(2) 记 $f(z) = \frac{1}{(z^2 + a^2)(z^2 + b^2)}$, 则

$$\lim_{z \rightarrow \infty} z f(z) = \lim_{z \rightarrow \infty} \frac{z}{(z^2 + a^2)(z^2 + b^2)} = 0,$$

那么

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{dx}{(x^2 + a^2)(x^2 + b^2)} &= 2\pi i \operatorname{Res}[f(z), ai] + 2\pi i \operatorname{Res}[f(z), bi] \\ &= 2\pi i \left[\frac{1}{2ai(b^2 - a^2)} + \frac{1}{2bi(a^2 - b^2)} \right] = \frac{\pi}{ab(a + b)}. \end{aligned}$$

(3) 记 $f(z) = \frac{1 + z^2}{1 + z^4}$, 则

$$\lim_{z \rightarrow \infty} z f(z) = \lim_{z \rightarrow \infty} \frac{z^3 + z^2}{z^4 + 1} = 0,$$

注意到 $f(x)$ ($x \in \mathbb{R}$) 为偶函数, 那么

$$\begin{aligned} \int_0^\infty \frac{1 + x^2}{1 + x^4} dx &= \frac{1}{2} \int_{-\infty}^{\infty} \frac{1 + x^2}{1 + x^4} dx = \pi i \operatorname{Res}[f(z), e^{\pi i/4}] + \pi i \operatorname{Res}[f(z), e^{3\pi i/4}] \\ &= \pi i \left[\frac{1 + i}{2\sqrt{2}(-1 + i)} + \frac{1 - i}{2\sqrt{2}(1 + i)} \right] = \frac{\pi i}{2\sqrt{2}} (e^{-\pi i/2} + e^{-\pi i/2}) = \frac{\pi}{\sqrt{2}}. \end{aligned}$$

6. (1) 引入辅助函数 $f(z) = \frac{z}{z^2 + b^2} e^{iaz}$, 其中 $(z^2 + b^2)$ 比 z 高 1 次. 又因为被积函数为偶函数, 那么

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \pi i \operatorname{Res}[f(z), bi] = \pi i \cdot \frac{bi}{2bi} e^{-ab} = \frac{\pi}{2b} e^{-abi}, \\ \int_0^\infty \frac{x \sin ax}{x^2 + b^2} dx &= \frac{1}{2} \operatorname{Im} \left[\int_{-\infty}^{\infty} f(z) dz \right] = \frac{\pi}{2b} e^{-ab}. \end{aligned}$$

- (2) 引入辅助函数 $f(z) = \frac{e^{iaz}}{z(z^2 + b^2)}$, 其中 $z(z^2 + b^2)$ 比 1 高 3 次. 又因为被积函数为偶函数, 那么

$$\begin{aligned}\int_{-\infty}^{\infty} f(x) dx &= 2\pi i \operatorname{Res}[f(z), bi] + \pi i \operatorname{Res}[f(z), 0] \\ &= 2\pi i \cdot \frac{e^{-ab}}{bi \cdot 2bi} + \pi i \cdot \frac{1}{b^2} = \frac{\pi i}{b^2} (1 - e^{-ab}), \\ \int_0^{\infty} \frac{\sin ax}{x(x^2 + b^2)} dx &= \frac{1}{2} \operatorname{Im} \left[\int_{-\infty}^{\infty} f(x) dx \right] = \frac{\pi}{2b^2} (1 - e^{-ab}).\end{aligned}$$

- (3) 引入辅助函数 $f(z) = \frac{(z^2 - a^2)e^{iz}}{(z^2 + a^2)z}$, 其中 $z(z^2 + a^2)$ 比 $(z^2 - a^2)$ 高 1 次. 又因为被积函数为偶函数, 那么

$$\begin{aligned}\int_{-\infty}^{\infty} f(x) dx &= 2\pi i \operatorname{Res}[f(z), ai] + \pi i \operatorname{Res}[f(z), 0] \\ &= 2\pi i \cdot \frac{-2a^2 \cdot e^{-a}}{2ai \cdot ai} + \pi i \cdot (-1) = \pi(2e^{-a} - 1)i, \\ \int_0^{\infty} \frac{x^2 - a^2}{x^2 + a^2} \frac{\sin x}{x} dx &= \frac{1}{2} \operatorname{Im} \left[\int_{-\infty}^{\infty} f(x) dx \right] = \pi \left(e^{-a} - \frac{1}{2} \right).\end{aligned}$$

- (4) 引入辅助函数 $f(z) = \frac{e^{iz}}{(z^2 + 4)(z - 1)}$, 其中 $(z^2 + 4)(z - 1)$ 比 1 高 3 次, 那么

$$\begin{aligned}\int_{-\infty}^{\infty} f(x) dx &= 2\pi i \operatorname{Res}[f(z), 2i] + \pi i \operatorname{Res}[f(z), 1] \\ &= 2\pi i \cdot \frac{e^{-2}}{4i(2i - 1)} + \pi i \cdot \frac{e^i}{5} \\ &= \frac{\pi}{10} [-(2 \sin 1 + e^{-2}) + 2(\cos 1 - e^{-2})], \\ \text{v.p.} \int_{-\infty}^{\infty} \frac{\sin x}{(x^2 + 4)(x - 1)} dx &= \operatorname{Im} \left[\int_{-\infty}^{\infty} f(x) dx \right] = \frac{\pi}{5} (\cos 1 - e^{-2}).\end{aligned}$$

- (5) 引入辅助函数 $f(z) = \frac{e^{2azi} - e^{2bzi}}{z^2}$, 其中 z^2 比 1 高 2 次. 又因为被积函数是偶函数, 那么

$$\begin{aligned}\int_{-\infty}^{\infty} f(x) dx &= \pi i \lim_{z \rightarrow 0} \frac{d}{dz} (e^{2azi} - e^{2bzi}) = 2\pi(b - a), \\ \int_0^{\infty} \frac{\cos 2ax - \cos 2bx}{x^2} dx &= \frac{1}{2} \operatorname{Re} \left[\int_{-\infty}^{\infty} f(x) dx \right] = \pi(b - a).\end{aligned}$$

- (6) 将原积分改写为

$$\int_0^{\infty} \left(\frac{\sin x}{x} \right)^2 dx = \int_0^{\infty} \frac{1 - \cos 2x}{2x^2} dx,$$

故引入辅助函数 $f(z) = \frac{1 - e^{2iz}}{z^2}$, 其中 z^2 比 1 高 2 次. 又因为被积函数是偶函数,

那么

$$\begin{aligned}\int_{-\infty}^{\infty} f(x) dx &= \pi i \operatorname{Res}[f(z), 0] = \pi i \lim_{z \rightarrow 0} \frac{1 - e^{2iz}}{2z} \\ &= -\frac{\pi i}{2} \lim_{z \rightarrow 0} \sum_{n=1}^{\infty} (2i)^n z^{n-1} = \pi, \\ \int_0^{\infty} \left(\frac{\sin x}{x} \right)^2 dx &= \frac{1}{2} \operatorname{Re} \left[\int_{-\infty}^{\infty} f(x) dx \right] = \frac{\pi}{2}.\end{aligned}$$

注 对于函数 $f(z) = \frac{1 - e^{2iz}}{2z^2}$ 在 C_R 上的积分, 可令 $f_1(z) = \frac{1}{z^2}$. 由于

$$\lim_{z \rightarrow \infty} z f_1(z) = \lim_{z \rightarrow \infty} \frac{1}{z} = 0, \quad \lim_{z \rightarrow \infty} f_1(z) = \lim_{z \rightarrow \infty} \frac{1}{z^2} = 0,$$

分别由引理 1 和约当引理, 可得

$$\lim_{R \rightarrow \infty} \int_{C_R} f_1(z) dz = 0, \quad \lim_{R \rightarrow +\infty} \int_{C_R} f_1(z) e^{2iz} dz = 0.$$

(7) 利用提示, 引入辅助函数 $f(z) = \frac{z}{e^{\pi z} - e^{-\pi z}}$. 注意到

$$\begin{aligned}\lim_{z \rightarrow 0} f(z) &= \lim_{z \rightarrow 0} \left[\sum_{n=0}^{\infty} \frac{\pi^n z^{n-1}}{n!} - \sum_{n=0}^{\infty} \frac{(-\pi)^n z^{n-1}}{n!} \right]^{-1} \\ &= \lim_{z \rightarrow 0} \left[\sum_{n=0}^{\infty} \frac{2\pi^{2n+1} z^{2n}}{(2n+1)!} \right]^{-1} = \frac{1}{2\pi},\end{aligned}$$

因此 $z = 0$ 实际上为 $f(z)$ 的可去奇点.

考虑由 $z = 0, z = R, z = ih$ ($h \in \mathbb{R}$), $z = -R$ 围成的正向复闭路, 上述四条路线依次记为 C_1, C_2, C_3, C_4 , 那么

$$\int_C f(z) dz = \int_{-\infty}^{\infty} f(x) dx + \int_{C_2} f(z) dz + \int_{C_3} f(z) dz + \int_{C_4} f(z) dz.$$

当 z 在 C_2 上时, $z = R + iy$, 由长大不等式,

$$\begin{aligned}\left| \int_{C_2} f(z) dz \right| &\leq \int_0^h \frac{|z|}{|e^{\pi z} - e^{-\pi z}|} |dz| \leq \int_0^h \frac{\sqrt{R^2 + h^2}}{e^{\pi R} - e^{-\pi R}} dy \\ &\leq \frac{h\sqrt{R^2 + h^2}}{e^{\pi R}} \rightarrow 0 \quad (R \rightarrow \infty),\end{aligned}$$

因此 $f(z)$ 在 C_2 上的积分值当 $R \rightarrow \infty$ 时为零. 同理, $f(z)$ 在 C_4 上的积分值当 $R \rightarrow \infty$ 时同样为零.

对于 $f(z)$ 在 C_3 上的积分, 为使计算方便, 取 $h = 1/2$, 那么 $z = x + i/2$,

$$\begin{aligned}\int_{-R}^R f(x) dx &= \int_{-R}^R f(z) dz = \int_{-R}^R \frac{x + i/2}{ie^{\pi x} - (-ie^{-\pi x})} dx = \int_{-R}^R \frac{1 - 2ix}{2(e^{\pi x} + e^{-\pi x})} dx \\ &= \frac{1}{2} \int_{-R}^R \frac{dx}{e^{\pi x} + e^{-\pi x}} = \frac{1}{4\pi} \int_{-R}^R \frac{d(\sinh \pi x)}{\cosh^2 \pi x} = \frac{1}{4\pi} \arctan(\sinh \pi x) \Big|_{-R}^R \\ &= \frac{1}{2\pi} \arctan(\sinh \pi R),\end{aligned}$$

综上, 积分值为

$$\begin{aligned}\int_0^\infty \frac{x}{e^{\pi x} - e^{-\pi x}} dx &= \frac{1}{2} \lim_{R \rightarrow \infty} \int_{-R}^R f(x) dx = \frac{1}{4\pi} \lim_{R \rightarrow \infty} \arctan(\sinh \pi R) \\ &= \frac{1}{4\pi} \cdot \frac{\pi}{2} = \frac{1}{8}.\end{aligned}$$

7. (1) 引入辅助函数 $R(z) = \frac{1}{(1+z^2)^2}$, 可见 $R(x)$ 在正实轴上无奇点, 且

$$\begin{aligned}\lim_{z \rightarrow \infty} z^{p+1} R(z) &= \lim_{z \rightarrow \infty} \frac{z^{p+1}}{(1+z^2)^2} = \lim_{z \rightarrow \infty} \frac{1}{z^{3-p}} = 0, \\ \lim_{z \rightarrow 0} z^{p+1} R(z) &= \lim_{z \rightarrow 0} \frac{z^{p+1}}{(1+z^2)^2} = \lim_{z \rightarrow 0} \frac{0}{1^2} = 0,\end{aligned}$$

那么

$$\begin{aligned}\int_0^\infty x^p R(x) dx &= \frac{2\pi i}{1 - e^{2p\pi i}} \{ \text{Res}[z^p R(z), i] + \text{Res}[z^p R(z), -i] \} \\ &= \frac{2\pi i}{1 - e^{2p\pi i}} \left[\frac{1-p}{4} e^{(p-1)\pi i/2} + \frac{1-p}{4} e^{3(p-1)\pi i/2} \right] \quad (0 \leq \theta < 2\pi) \\ &= \frac{(1-p)\pi}{2(e^{2p\pi i} - 1)} \cdot (e^{3p\pi i/2} - e^{p\pi i/2}) = \frac{(1-p)\pi}{2(e^{p\pi i} - e^{-p\pi i})} \cdot (e^{p\pi i/2} - e^{-p\pi i/2}) \\ &= \frac{(1-p)\pi \sin \frac{p\pi}{2}}{2 \sin p\pi} = \frac{(1-p)\pi}{4 \cos \frac{p\pi}{2}}.\end{aligned}$$

并且

$$\lim_{p \rightarrow 1} \int_0^\infty x^p R(x) dx \stackrel{\text{L'Hospital 法则}}{=} \lim_{p \rightarrow 1} \frac{\pi}{2\pi \sin \frac{p\pi}{2}} = \frac{1}{2},$$

因此

$$\int_0^\infty x^p R(x) dx = \begin{cases} \frac{(1-p)\pi}{4 \cos \frac{p\pi}{2}}, & p \neq 1, \\ \frac{1}{2}, & p = 1 \end{cases} \quad (-1 < p < 3).$$

(2) 记 $f(z) = \frac{(\ln z)^2}{(1+z)^3}$. 注意到 $\ln z$ 的多值性, 取正实轴为支割线, 在上岸 $f(z) = \frac{\ln^2 x}{(1+x^3)}$, 在下岸 $f(z) = \frac{\ln^2(xe^{2\pi i})}{(1+xe^{2\pi i})^3}$. 取闭路 C 为: 从正实轴上岸 $z = r$ 点到 $z = R$ 点, 再沿 $C_R: |Z| = R$ (正向) 从正实轴上岸到下岸, 接着从下岸 $Z = R$ 点到 $z = r$ 点, 最后沿 $-C_r: |z| = r$ (反向) 回到起点, 那么 $f(z)$ 在 C 围成的闭路内

⁸ 不能构造 $f(z) = \frac{\ln z}{(1+|z|)^3}$, 这样的 $f(z)$ 在半圆环带区域内并不解析, 也就是说, 从出发点就错了

仅有一个 3 级奇点 $z_0 = -1$, 从而

$$\begin{aligned}\int_C f(z) dz &= \int_{C_R} f(z) dz + \int_{-C_r} f(z) dz + \int_r^R \frac{\ln^2 x}{(1+x)^3} dx + \int_R^r \frac{\ln^2(xe^{2\pi i})}{(1+xe^{2\pi i})^3} dx \\ &= \int_{C_R} f(z) dz + \int_{-C_r} f(z) dz + \int_r^R \frac{-4\pi i \ln x + 4\pi^2}{(1+x)^3} dx \\ &= 2\pi i \operatorname{Res}[f(z), -1] = 2\pi i \cdot \frac{1}{2!} \lim_{z \rightarrow -1} \frac{d^2}{dz^2} (\ln^2 z) = 2\pi(\pi + i).\end{aligned}$$

由于

$$|\ln z| = \sqrt{(\ln |z|)^2 + \varphi^2} \leq \sqrt{\ln^2 |z| + 4\pi^2},$$

因此有

$$\begin{aligned}\left| \lim_{z \rightarrow \infty} z f(z) \right| &\leq \lim_{z \rightarrow \infty} \frac{|z| |\ln z|^2}{|1+z|^3} \leq \lim_{|z| \rightarrow \infty} \frac{|z| (\ln^2 |z| + 4\pi^2)}{(|z|-1)^3} = 0, \\ \left| \lim_{z \rightarrow 0} z f(z) \right| &\leq \lim_{z \rightarrow 0} \frac{|z| |\ln z|^2}{|1+z|^3} \leq \lim_{|z| \rightarrow 0} (|z| \ln^2 |z| + 4\pi^2 |z|) = 0,\end{aligned}$$

即 $\lim_{z \rightarrow \infty} z f(z) = \lim_{z \rightarrow 0} z f(z) = 0$, 由引理 1、2 得

$$\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = \lim_{r \rightarrow 0^+} \int_{C_r} f(z) dz = 0,$$

代入最初的方程并令 $R \rightarrow \infty, r \rightarrow 0$, 得

$$4\pi^2 \int_0^\infty \frac{dx}{(1+x)^3} - 4\pi i \int_0^\infty \frac{\ln x}{(1+x)^3} dx = 2\pi^2 + 2\pi i,$$

对比等式两端虚部得

$$\int_0^\infty \frac{\ln x}{(1+x)^3} dx = \frac{2\pi i}{-4\pi i} = -\frac{1}{2}.$$

- (3) 记 $f(z) = \frac{\ln^2 z}{z^2 + a^2}$, 取闭路 C 为: 从 $z = r$ 点到 $z = R$ 点, 再经 $C_R: |z| = R$ (正向) 到达 $z = -R$ 点. 然后从 $z = -R$ 点到 $z = -r$ 点, 最后经 $-C_r: |z| = r$ (反向) 回到起点, 且 $r < a < R$, 则 $f(z)$ 在闭路 C 所包围的区域内仅有一个 1 级极点 $z_0 = ai$, 那么

$$\begin{aligned}\int_C f(z) dz &= \int_{C_R} f(z) dz + \int_{-C_r} f(z) dz + \int_r^R \frac{\ln^2 x}{x^2 + a^2} dx + \int_{-R}^{-r} \frac{\ln^2(xe^{i\pi})}{(xe^{i\pi})^2 + a^2} d(xe^{i\pi}) \\ &= \int_{C_R} f(z) dz + \int_{-C_r} f(z) dz + \int_r^R \frac{2\ln^2 x - \pi^2}{x^2 + a^2} dx + 2\pi i \int_0^\infty \frac{\ln x}{x^2 + a^2} dx.\end{aligned}$$

注意到

$$\begin{aligned}\left| \lim_{z \rightarrow \infty} z f(z) \right| &\leq \lim_{z \rightarrow \infty} \frac{|z| |\ln z|^2}{|z^2 + a^2|} \leq \lim_{|z| \rightarrow \infty} \frac{|z| (\ln^2 |z| + \pi^2)}{|z|^2 - a^2} = 0, \\ \left| \lim_{z \rightarrow 0} z f(z) \right| &\leq \lim_{z \rightarrow 0} \frac{|z| |\ln z|^2}{|z^2 + a^2|} = \frac{1}{a^2} \lim_{|z| \rightarrow 0} |z| (\ln^2 |z| + \pi^2) = 0,\end{aligned}$$

即 $\lim_{z \rightarrow \infty} z f(z) = \lim_{z \rightarrow 0} z f(z) = 0$, 由引理 1、2 得

$$\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0, \quad \lim_{r \rightarrow 0^+} \int_{C_r} f(z) dz,$$

代入最初的方程并令 $R \rightarrow \infty, r \rightarrow 0^+$, 得

$$\begin{aligned} \int_C f(z) dz &= 2 \int_0^\infty \frac{\ln^2 x}{x^2 + a^2} dx - \pi^2 \cdot \frac{\pi}{2|a|} - 2\pi i \int_0^\infty \frac{\ln x}{x^2 + a^2} dx \\ &= 2\pi i \operatorname{Res}[f(z), ai] = 2\pi i \cdot \frac{\ln^2(|a|i)}{2|a|i} = \frac{\pi}{|a|} \left(\ln^2 |a| - \frac{\pi^2}{4} + i\pi \ln |a| \right), \end{aligned}$$

对比等式两端实部得

$$\int_0^\infty \frac{\ln^2 x}{x^2 + a^2} dx = \frac{1}{2} \cdot \frac{\pi}{|a|} \left(\ln^2 |a| + \frac{\pi^2}{4} \right) = \frac{\pi}{8|a|} (\pi^2 + \ln^2 |a|).$$

8. (1) 记 $f(z) = 8, \varphi(z) = 2z^5 - z^4 + z^2 - 2z$, 那么在 $|z| = 1$ 上

$$|\varphi(z)| \leq 2|z|^5 + |z|^4 + |z|^2 + 2|z| = 6 < |f(z)| = 8,$$

而 $f(z)$ 在 $|z| < 1$ 内无零点, 由 Rouché 定理, $f(z) + \varphi(z)$ 在 $|z| < 1$ 内无零点.

(2) 记 $f(z) = -6z^5, \varphi(z) = z^7 + z^2 - 3$, 那么在 $|z| = 1$ 上

$$|\varphi(z)| \leq 1 + 1 + 3 = 5 < |f(z)| = 6,$$

而 $f(z)$ 在 $|z| < 1$ 内仅有一个 5 级零点 $z_0 = 0$, 由 Rouché 定理, $f(z) + \varphi(z)$ 在 $|z| < 1$ 内有 5 个零点.

(3) 记 $f(z) = -8z, \varphi(z) = z^9 - 2z^6 + z^2 + 2$, 那么在 $|z| = 1$ 上

$$|\varphi(z)| \leq 1 + 2 + 1 + 2 = 6 < |f(z)| = 8,$$

而 $f(z)$ 在 $|z| < 1$ 内仅有一个 1 级零点 $z_0 = 0$, 由 Rouché 定理, $f(z) + \varphi(z)$ 在 $|z| < 1$ 内仅有一个零点.

(4) 记 $f(z) = -3z^n, \varphi(z) = e^z$, 那么在 $|z| = 1$ 上

$$|\varphi(z)| = \left| \sum_{n=0}^{\infty} \frac{z^n}{n!} \right| \leq \sum_{n=0}^{\infty} \frac{|z|^n}{n!} = \sum_{n=0}^{\infty} \frac{1}{n!} = e < |f(z)| = 3,$$

而 $f(z)$ 在 $|z| < 1$ 内仅有一个 n 级零点, 由 Rouché 定理, $f(z) + \varphi(z)$ 在 $|z| < 1$ 内有 n 个零点.

9. 将区域分成两部分讨论:

(1) 在 $|z| = \frac{1}{2}$ 上, 令 $f(z) = 6z, \varphi(z) = z^4 + 1$, 则

$$|\varphi(z)| \leq |z|^4 + 1 = 1 + \frac{1}{16} < |f(z)| = 6|z| = 3,$$

而 $f(z)$ 在 $|z| < \frac{1}{2}$ 内仅有一个 1 级零点 $z_0 = 0$, 由 Rouché 定理, $f(z) + \varphi(z)$ 在 $|z| < \frac{1}{2}$ 内仅有 1 个零点.

在 $|z| = 2$ 上, 令 $f(z) = z^4, \varphi(z) = 6z + 1$, 则

$$|\varphi(z)| \leq 6|z| + 1 = 13 < |f(z)| = |z|^4 = 16,$$

而 $f(z)$ 在 $|z| < 2$ 内仅有一个 4 级零点 $z_0 = 0$, 由 Rouché 定理, $f(z) + \varphi(z)$ 在 $|z| < 2$ 内有 3 个零点.

综上, $z^4 + 6z + 1 = 0$ 在 $\frac{1}{2} < |z| < 2$ 内共有 $4 - 1 = 3$ 个根.

- (2) 将方程改写 $(z - \lambda)e^z + 1 = 0$, 则记 $f(z) = (z - \lambda)e^z, \varphi(z) = 1$. 对于足够大的 $R \in \mathbb{R}$, 取闭路 C 为: 从 $z = -iR$ 沿 $|z| = R$ 逆时针到达 $z = iR$ 点, 再从 $z = iR$ 沿虚轴返回起点.

在 $|z| = R$ ($-\pi/2 < \theta < \pi/2$) 上,

$$|f(z)| = |z - \lambda||e^z| \geq (|z| - \lambda)e^x \geq R - \lambda > 1 = |\varphi(z)|,$$

在虚轴上, $z = iy$ ($-R < y < R$), 那么

$$|f(z)| = |iy - \lambda||e^{iy}| = \sqrt{\lambda^2 + y^2} \geq \lambda > 1 = |\varphi(z)|,$$

即在 C 上总有 $|f(z)| > |\varphi(z)|$, 而 $f(z)$ 在 C 所围成的区域内仅有一个 1 级零点 $z_0 = \lambda$, 由 Rouché 定理, 令 $R \rightarrow \infty$, $f(z) + \varphi(z) = 0$ 在右半平面内仅有一个零点. 对于函数 $g(x) = \lambda - x - e^{-x}$, 考虑到

$$g(0) = \lambda - 1 > 0, \quad g(\lambda) = -e^{-\lambda} < 0,$$

由零点存在定理, $\exists \xi \in (0, \lambda)$, 使得 $g(\xi) = 0$. 可见 $g(z) = 0$ 在实轴上有零点. 由零点的唯一性, $g(z) = 0$ 在右半平面内唯一的一个根为此实根 $z = \xi$.

10. 记多项式函数 $f(z) = z^4 + z^3 + 4z^2 + 2z + 3$, 则 $n = 4$. 现研究

$$f(iy) = y^4 - iy^3 - 4y^2 + 2iy + 3 = (y^4 - 4y^2 + 3) + (-y^3 + 2y)i \triangleq u(y) + iv(y),$$

其中 $u(y)$ 零点为 $\pm 1, \pm \sqrt{3}$, $v(y)$ 零点为 $0, \pm \sqrt{2}$. 当 y 从 $-\infty$ 到 $+\infty$ 时,

$$u'(y) = 4y^3 - 8y = 4y(y^2 - 2), \quad v'(y) = -3y^2 + 2,$$

$u(y), v(y)$ 的变化趋势如下表: 可见 $f(z)$ 在 y 从 $-\infty$ 到 ∞ 的过程中绕原点转了 $k = 2$

y	$-\infty$	$-\sqrt{3}^-$	$-\sqrt{2}^-$	-1^-	0^-	1^-	$\sqrt{2}^-$	$\sqrt{3}^-$	∞
$u(y)$	$\infty \uparrow$	$0 \downarrow$	$-1 \uparrow$	$0 \uparrow$	$3 \downarrow$	$0 \downarrow$	$-1 \uparrow$	$0 \uparrow$	$\infty \uparrow$
$v(y)$	$\infty \downarrow$	$\sqrt{3} \downarrow$	$0 \downarrow$	$-1 \downarrow$	$0 \uparrow$	$1 \downarrow$	$0 \downarrow$	$-\sqrt{3} \downarrow$	$\infty \downarrow$
$v(y)/u(y)$	0	∞	0	∞	0	∞	0	∞	0
$\arg f(z)$	0	$\pi/2$	π	$3\pi/2$	0	$\pi/2$	π	$3\pi/2$	0

圈, 因此 $f(z)$ 在左半平面共有 $\left(\frac{n}{2} + k\right) = 4$ 个零点.

6 解析开拓

1. 与唯一性定理并不矛盾. 记 $f(z) = \sin \frac{1}{1-z}$, $g(z) \equiv 0$, 由题意可知,

$$f\left(1 - \frac{1}{k\pi}\right) = g\left(1 - \frac{1}{k\pi}\right) = 0, \quad k = 1, 2, \dots,$$

但 $\lim_{k \rightarrow \infty} \left(1 - \frac{1}{k\pi}\right) = 1$, 此极限值并不在 $|z| < 1$ 内, 因此无法使用唯一性定理证明 $f(z) \equiv g(z)$.

2. (1) a

(2) 假定存在函数 $f(z)$, 使得

$$f\left(\frac{1}{n}\right) = \begin{cases} 0, & n = 2k-1, \\ \frac{1}{2k}, & n = 2k \end{cases} \quad (k \in \mathbb{N}^*),$$

记 $g(z) = \frac{1}{z}$ ($z \neq 0$), 则 $f\left(\frac{1}{2k}\right) = g\left(\frac{1}{2k}\right)$, 但极限值 $\lim_{k \rightarrow \infty} \frac{1}{2k} = 0$ 并不在 $g(z)$ 的解析域上, 因此无法利用唯一性定理证明 $f(z) \equiv g(z)$, 即假设不成立, 不存在这样的函数 $f(z)$.

- (3) a

- (4) 取 $z_n = \frac{1}{n}$, $g(z) = \frac{1}{1+z}$ ($z \neq -1$). 假设存在这样的 $f(z)$, 使得

$$f\left(\frac{1}{n}\right) = \frac{n}{n+1} = \frac{1}{1 + \frac{1}{n}} = g\left(\frac{1}{n}\right),$$

并且极限值 $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ 在 $g(z)$ 的解析域内, 由唯一性定理, $f(z) \equiv g(z) = \frac{1}{1+z}$ ($z \neq -1$).

7 保形变换及其应用

- 转动角 $\arg w'(1) = \arg 3 = 0$, 伸张系数 $|w'(1)| = 3$.
 - 转动角 $\arg w'\left(\frac{1}{2}\right) = \arg \frac{3}{4} = 0$, 伸张系数 $\left|w'\left(\frac{1}{2}\right)\right| = \frac{3}{4}$.
 - 转动角 $\arg w'(1+i) = \arg 6i = \frac{\pi}{2}$, 伸张系数 $|w'(1+i)| = 6$.
 - 转动角 $\arg w'(\sqrt{3}-i) = \arg 6(1-\sqrt{3}i) = -\frac{\pi}{3}$, 伸张系数 $|w'(\sqrt{3}-i)| = 12$.
- 伸张系数 $|w'(z)| = 2|z|$, 则 $|z| > \frac{1}{2}$ 的部分被放大, $|z| < \frac{1}{2}$ 的部分被缩小.
 - 伸张系数 $|w'(z)| = \frac{1}{|z|^2}$, 则 $|z| < 1$ 的部分被放大, $|z| > 1$ 的部分被缩小.
 - 伸张系数 $|w'(z)| = |e^z| = e^x$, 则 $x > 0$ 的部分被放大, $x < 0$ 的部分被缩小.
- 设 $z(t) = x(t) + iy(t)$, 那么 L 的长度为

$$\int_{\alpha}^{\beta} \sqrt{x'^2(t) + y'^2(t)} dt = \int_{\alpha}^{\beta} |z'(t)| dt.$$

- 结合上一问结论 L' 的长度为

$$\int_{\alpha}^{\beta} \left| \frac{df[z(t)]}{dt} \right| dt = \int_{\alpha}^{\beta} |f'[z(t)]z'(t)| dt.$$

- 设 $w = f(z) = u(x, y) + iv(x, y)$, 那么由 C-R 方程,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}, \quad f'(z) = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}i,$$

因此区域 G 的面积为

$$\begin{aligned} \iint_G du dv &= \iint_D \left\| \frac{\partial(u, v)}{\partial(x, y)} \right\| dx dy = \iint_D \left| \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right| dx dy \\ &= \iint_D \left| \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right| dx dy = \iint_D |f'(z)|^2 dx dy. \end{aligned}$$

当 $w = z^2$ 且积分区域 $D = \{(x, y) | 0 \leq x, y \leq 1\}$ 时, 区域 G 的面积为

$$\begin{aligned} \iint_D |f'(z)|^2 dx dy &= \iint_D 4|z|^2 dx dy = 4 \int_0^1 dx \int_0^1 (x^2 + y^2) dy \\ &= 4 \int_0^1 \left(x^2 + \frac{1}{3} \right) dx = \frac{8}{3}. \end{aligned}$$

- 设各圆周方程为 $C_i: |z - a_i| = r_i$ ($r_i > 0$), 且 C_1, C_2 相切于原点. 对这样的圆, 总可以选取实轴方向使得 C_1, C_2 的圆心在实轴上, 且 C_3 位于上半平面. 不妨设 $r_2 \geq r_1$, 则规定实轴方向为从 C_2 圆心指向 C_1 圆心, 那么两者方程简化为

$$C_1: |z - r_1| = r_1, \quad C_2: |z + r_2| = r_2,$$

对于 C_1 , 代入变换 $w = \frac{1}{z}$ 得

$$r_1^2 = |z - r_1|^2 = z\bar{z} - r_1(z + \bar{z}) + r_1^2 = \frac{1}{w\bar{w}} - r_1\left(\frac{1}{w} + \frac{1}{\bar{w}}\right) + r_1^2,$$

整理得 $r_1(w + \bar{w}) = 1$, 即 $C'_1: \operatorname{Re}(w) = \frac{1}{2r_1}$. 类似地, C_2 经变换得 $C'_2: \operatorname{Re}(w) = \frac{1}{-2r_2}$.

注意到 $w(r_1) = \frac{1}{r_1} > \frac{1}{2r_1}$, 因此 C_1, C_2 的圆内域分别变为区域 $\operatorname{Re}(w) \leq -\frac{1}{2r_2}$, $\operatorname{Re}(w) \geq \frac{1}{2r_1}$.

对于 C_3 , 其圆心满足方程

$$\begin{cases} |a_3 - r_1| = r_1 + r_3, \\ |a_3 + r_2| = r_2 + r_3, \end{cases}$$

设 $a_3 = x_3 + iy_3$, 代入上述方程组解得

$$a_3 = x_3 + iy_3 = \frac{r_2 - r_1}{r_2 + r_1}r_3 + \frac{2i}{r_2 + r_1}\sqrt{r_1r_2r_3(r_1 + r_2 + r_3)},$$

代入变换 $w = \frac{1}{z}$, C_3 的方程可写为 $\left|\frac{1}{w} - a_3\right| = r_3$. 令 $w = u + iv$, 可得

$$\begin{aligned} r_3^2|w|^2 &= |1 - a_3w|^2 = 1 - (a_3w + \bar{a}_3\bar{w}) + |a_3|^2|w|^2 = 1 - 2\operatorname{Re}(a_3w) + |a_3|^2|w|^2, \\ |a_3|^2 &= \left(\frac{r_2 - r_1}{r_2 + r_1}r_3\right)^2 + \left(\frac{2\sqrt{r_1r_2r_3(r_1 + r_2 + r_3)}}{r_2 + r_1}\right)^2 \\ &= \frac{r_3}{(r_2 + r_1)^2} [r_3(r_2 + r_1)^2 + 4r_1r_2(r_2 + r_1)] = r_3^2 + \frac{4r_1r_2r_3}{r_2 + r_1}, \\ \operatorname{Re}(a_3w) &= \frac{r_2 - r_1}{r_2 + r_1}r_3u - \frac{2v}{r_2 + r_1}\sqrt{r_1r_2r_3(r_1 + r_2 + r_3)}, \end{aligned}$$

因此整理得 C_3 经变换得 C'_3 的方程:

$$\left(u - \frac{r_2 - r_1}{4r_2r_1}\right)^2 + \left(v + \sqrt{\frac{r_1 + r_2 + r_3}{4r_1r_2r_3}}\right)^2 = \frac{(r_2 + r_1)^2}{16r_2^2r_1^2}.$$

注意到当 $u = -\frac{1}{2r_2}$ 或 $u = \frac{1}{2r_1}$ 时, $v = 0$, 可见 C'_1, C'_2, C'_3 仍保持相切. 又因为待求区域为圆外域交集中不含 ∞ 点的区域, 而 $w(\infty) = 0$, 并且 $w = 0$ 的点在 C'_3 所包围的圆的外部区域. 因此 C_3 的圆内域变换为 C'_3 的圆内域:

$$\left|w - \left(\frac{r_2 - r_1}{4r_2r_1} - i\sqrt{\frac{r_1 + r_2 + r_3}{4r_1r_2r_3}}\right)\right| \leq \frac{r_2 + r_1}{4r_2r_1}.$$

综上, 这三个圆的边界围成的封闭图形的外部区域即三个圆外域的交集中包含 ∞ 点的连续区域. 因此, 在 $w = \frac{1}{z}$ 的变换下, 其形状为一半无限长且端口为一内凹的半圆的区域, 在此坐标轴选取下, 其可表示为

$$-\frac{1}{2r_2} \leq \operatorname{Re}(w) \leq \frac{1}{2r_1}, \quad \operatorname{Im}(w) \leq \sqrt{\frac{(r_2 + r_1)^2}{16r_2^2r_1^2} - \left(\operatorname{Re}(w) - \frac{r_2 - r_1}{4r_2r_1}\right)^2} - \sqrt{\frac{r_1 + r_2 + r_3}{4r_1r_2r_3}}.$$

6. 利用公式

$$\frac{w - w_1}{w - w_2} \frac{w_3 - w_2}{w_3 - w_1} = \frac{z - z_1}{z - z_2} \frac{z_3 - z_2}{z_3 - z_1}.$$

(1) 取 $z_1 = -1, z_2 = \infty, z_3 = i$, 那么方程变为

$$\frac{w - i}{w - 1} \frac{1}{1} = \frac{z + 1}{1} \frac{1}{1 + i},$$

从中解得分式线性变换 M 为

$$w = \frac{z + 2 + i}{z + 2 - i}.$$

(2) 取 $z_1 = \infty, z_2 = i, z_3 = -1$, 那么方程变为

$$\frac{w - i}{w - 1} \frac{1}{1} = \frac{1}{z - i} \frac{-1 - i}{1},$$

从中解得

$$w = \frac{iz + 2 + i}{z + 1}.$$

(3) 取 $z_1 = -1, z_2 = i, z_3 = \infty$, 那么方程变为

$$\frac{w}{w - 1} \frac{1}{1} = \frac{z + 1}{z - i} \frac{1}{1},$$

从中解得

$$w = \frac{z + 1}{1 + i} = \frac{1 - i}{2}(z + 1).$$

7. z 平面内区域 $\text{Im}z > 0$ 边界为实轴 $\text{Im}z = 0$, 显然 $-i$ 为 i 关于边界的对称点, 且 $w(i) = 0$, 因此 $w(-i) = \infty$, 变换可写为

$$w = k \frac{z - i}{z + i}.$$

而在变换后得到的圆的边界上,

$$1 = |w| = |k| \left| \frac{z - i}{z + i} \right| = |k| \left| \frac{x - i}{x + i} \right| = |k| \left| \frac{x - i}{\overline{x - i}} \right| = |k|,$$

因此 $k = e^{i\theta}$ ($0 \leq \theta < 2\pi$), 即 $w = e^{i\theta} \frac{z - i}{z + i}$. 又因为

$$w'(i) = e^{i\theta} \frac{2i}{(z + i)^2} \Big|_{z=i} = \frac{e^{i\theta}}{2i},$$

$$\arg w'(i) = \theta - \frac{\pi}{2} = -\frac{\pi}{2},$$

从中解得 $\theta = 0$, 因此最终得到线性变换 $w = w(z) = \frac{z - i}{z + i}$.

8. 由题意, $z_0 = \frac{1}{2}$ 关于圆 $|z| = 1$ 的对称点位于正实轴上, 且有模长关系: $|z'_0| \cdot \frac{1}{2} = 1^2$, 即 $z'_0 = 2$. 而在 w 平面上, 0 关于圆 $|w| = 1$ 的对称点为 ∞ , 因此

$$w\left(\frac{1}{2}\right) = 0, \quad w(2) = \infty,$$

那么分式线性变换可写为 $w(z) = k \frac{2z-1}{2(z-2)}$, 且要求当 $|z| = 1$ 时,

$$|w| = |k||z| \left| \frac{2-\bar{z}}{2(z-2)} \right| = |k| \cdot 1 \cdot \frac{1}{2} = 1,$$

因此 $k = 2e^{i\theta}$ ($0 \leq \theta < 2\pi$). 又因为

$$\arg w'\left(\frac{1}{2}\right) = \arg \frac{-3k}{(z-2)^2} \Big|_{z=1/2} = \arg \left(-\frac{4}{3}e^{i\theta}\right) = \theta - \pi = 0,$$

因此 $\theta = \pi$, 即 $k = -2$, 分式线性变换为 $w = -\frac{2z-1}{z-2}$.

9. 由题意, $w(a) = \bar{a}$, 那么 $w(\bar{a}) = a$, 因此设分式线性变换为

$$\frac{w - \bar{a}}{w - a} = k \frac{z - a}{z - \bar{a}},$$

注意到 $\operatorname{Im}(z) = 0$ 时, $\operatorname{Im}(w) = 0$, 即 $z = \bar{z}, \bar{w} = w$, 因此

$$\begin{aligned} \left| \frac{w - \bar{a}}{w - a} \right| &= |k| \left| \frac{z - a}{z - \bar{a}} \right| = |k| \left| \frac{z - z}{\bar{z} - \bar{a}} \right| = |k| \\ &= \left| \frac{w - \bar{a}}{\bar{w} - a} \right| = 1, \end{aligned}$$

因此 $k = e^{i\theta}$ ($-\pi \leq \theta < \pi$). 对上述分式线性变换两端关于 z 求导, 得

$$-w'(z) \cdot \frac{a - \bar{a}}{(w - a)^2} = k \frac{a - \bar{a}}{(z - \bar{a})^2}.$$

(1) 当 $a \in \mathbb{R}$ 时, $k = 1$,

(2) 当 $a \notin \mathbb{R}$ 时,

$$w'(a) = -k \frac{(w - a)^2}{(z - a)^2} \Big|_{z=a} = -e^{i\theta} \frac{(\bar{a} - a)^2}{(a - \bar{a})^2} = e^{i(\theta - \pi)},$$

因此 $\arg w'(a) = \theta - \pi = -\frac{\pi}{2}$, 即 $\theta = \frac{\pi}{2}$, 分式线性变换为

$$\frac{w(z) - \bar{a}}{w(z) - a} = i \frac{z - a}{z - \bar{a}} \quad \text{或} \quad w(z) = \frac{(\bar{a} - ai)z + (a^2i - \bar{a}^2)}{(1 - i)z + (ai - \bar{a})}.$$

10. (1) 在边界上, $\operatorname{Im}(z) = 1, |z| = \sqrt{(\operatorname{Re}(z))^2 + (\operatorname{Im}(z))^2} = 2$, 因此 $z_1 = \sqrt{3} - i$ 或 $z_2 = \sqrt{3} + i$. 再考虑 $z_3 = 2i$. 这里希望经变换 $t = t(z)$ 后,

$$t(-\sqrt{3} + i) = 0, \quad t(2i) = \infty, \quad t(\sqrt{3} + i) \in \mathbb{R},$$

结合前两式, 可得变换 $t(z) = k \frac{z - (-\sqrt{3} + i)}{z - 2i}$. 取 $|k| = 1$, 且

$$t(\sqrt{3} + i) = k \frac{2\sqrt{3}}{\sqrt{3} - i} = k \frac{3 + \sqrt{3}i}{2} \in \mathbb{R},$$

那么 $k = \frac{\sqrt{3} - i}{2}$, 由线性变换的保角性, 经变换

$$t(z) = \frac{\sqrt{3} - i}{2} \frac{z - (-\sqrt{3} + i)}{z - 2i}$$

后得到的区域是角域 $0 < \arg t < \frac{\pi}{3}$, 因此通过指数变换

$$\begin{aligned} w(z) &= [t(z)]^3 = \left(\frac{\sqrt{3} - i}{2} \frac{z + \sqrt{3} - i}{z - 2i} \right)^3 = e^{-\pi i/2} \left(\frac{z + \sqrt{3} - i}{z - 2i} \right)^3 \\ &= -i \left(\frac{z + \sqrt{3} - i}{z - 2i} \right)^3. \end{aligned}$$

(2) a

(3) 两个圆域边界交点分别为 $z_1 = \frac{1-i}{\sqrt{2}}, z_2 = \frac{-1-i}{\sqrt{2}}$, 两圆周在该点交角恰为 $\frac{\pi}{2}$, 那么作变换

$$t(z) = \frac{z - z_1}{z - z_2} = \frac{\sqrt{2}z - (1-i)}{\sqrt{2}z + (1+i)},$$

此时 z_1 变为 0, z_2 变为 ∞ , 由保角性, 其形成一个角域 $\alpha < \arg t < \alpha + \frac{\pi}{2}$. 考虑圆周 $|z + i| = 1$ 上一点 $z = 0$ 变换为

$$t(0) = \frac{-1+i}{1+i} = i,$$

因此该段变换后变成射线 $\arg z = \frac{\pi}{2}$, 另一段变为射线 $\arg z = \pi$, 因此二次变换

$$u(z) = e^{-\pi i/2} t(z) = e^{-\pi i/2} \frac{\sqrt{2}z - 1 + i}{\sqrt{2}z + 1 + i}$$

将角域中辐角小的一边与实轴正向重合, 最后经变换

$$w(z) = [u(z)]^2 = e^{-\pi i} \left(\frac{\sqrt{2}z - 1 + i}{\sqrt{2}z + 1 + i} \right)^2 = - \left(\frac{\sqrt{2}z - 1 + i}{\sqrt{2}z + 1 + i} \right)^2.$$

得到上半平面.

(4) 两圆周与实轴的交点分别为 $z_1 = -2, z_2 = 4, z_3 = 2$, 其中 z_2 为其公共交点. 设作线性变换 $t = t(z)$ 后,

$$t_1 = t(z_1) = 0, \quad t_2 = t(z_2) = \infty,$$

因此可取线性变换为 $t(z) = \frac{z - z_1}{z - z_2} = \frac{z + 2}{z - 4}$, 则 z_3 变为 $t_3 = -2$. 由保角性, 射线 $\arg t = \arg t_3 = \pi$ 与从原点到 ∞ 的两条射线交角均为 $\frac{\pi}{2}$, 因此变换后的区域为角域

$$\frac{\pi}{2} < \arg t < \frac{3\pi}{2},$$

因此为得到上半平面, 作旋转变换

$$w(z) = e^{-\pi i/2} t(z) = -i \frac{z + 2}{z - 4}.$$

- (5) 考虑圆周与实轴的交点 $z_1 = 0, z_2 = 2, z_3 = -2$, 其中 z_2 为公共交点. 设线性变换 $t = t(z)$ 将 z_1, z_2 分别变为 $t_1 = 0, t_2 = \infty$, 因此可取线性变换

$$t(z) = \frac{z}{z - 2},$$

此变换将 z_3 变为 $t_3 = \frac{1}{2}$, 由保角性, 射线 $\arg t = \arg t_3 = 0$ 为平角角域平分线, 因此角域为

$$-\frac{\pi}{2} < \arg t < \frac{\pi}{2},$$

因此为得到上半平面, 作旋转变换

$$w(z) = e^{\pi i/2} \frac{z}{z - 2} = \frac{iz}{z - 2}.$$

11. (1) 先将该角域映照为上半平面, 可作变换 $t(z) = z^{\pi/\alpha}$. 然后经线性变换 $w(t)$ 得单位圆, 不妨设 $w(i) = 0$, 那么 $w(-i) = \infty$, 线形变换可取

$$w(z) = \frac{t(z) - i}{t(z) + i} = \frac{z^{\pi/\alpha} - i}{z^{\pi/\alpha} + i}.$$

(2) a

- (3) 先作变换 $t = z^2$ 将第一象限变为上半平面. 记 $z_1 = \sqrt{2}i, z_2 = 0, z_3 = 1$, 则在此变换下,

$$t_1 = t(z_1) = -2, \quad t_2 = t(z_2) = 0, \quad t_3 = t(z_3) = 1,$$

现考虑分式线性变换 $w = w(t)$, 使得上半平面变为上半平面, 且满足

$$w(t_1) = -1, \quad w(t_2) = 1, \quad w(t_3) = \infty,$$

由于 $t_i \in \mathbb{R}$, 不妨取 $w_i = w(t_i)$ ($i = 1, 2, 3$) 均位于实轴上, 因此分式线性变换为

$$\frac{w - (-1)}{w - 1} \cdot \frac{1}{1} = \frac{t - (-2)}{t - 0} \cdot \frac{1 - 0}{1 - (-2)},$$

整理可得

$$w(z) = -\frac{2t(z) + 1}{t(z) - 1} = -\frac{2z^2 + 1}{z^2 - 1}.$$

(4) 该区域为 $0 < \arg z < 2\pi, |z| < 1$, 先作变换 $t(z) = \sqrt{z}$, 将其变为上半单位圆域:

$$0 < \arg t < \pi, \quad |t| < 1$$

设线性变换 $u(t)$ 将 $t_1 = -1, t_2 = 1$ 分别映照为 $u_1 = 0, u_2 = \infty$, 则可取线性变换

$$u(t) = \frac{t(z) + 1}{t(z) - 1} = \frac{\sqrt{z} + 1}{\sqrt{z} - 1}.$$

此线性变换将 $t_3 = i$ 映照为 $u_3 = -i$, 由保角性, 此时角域为

$$\pi < \arg u < \frac{3\pi}{2},$$

因此最后作变换 $w(z)$ 得到上半平面:

$$w(z) = u^2(z) = \left(\frac{\sqrt{z} + 1}{\sqrt{z} - 1} \right)^2.$$

8 拉氏变换

1. (1) $L\left[\frac{1}{2}\sin 2t + \cos 3t\right] = \frac{1}{p^2 + 4} + \frac{p}{p^2 + 9}.$

(2) $L[e^{3t} - e^{-2t}] = \frac{1}{p-3} - \frac{1}{p+2} = \frac{5}{(p-3)(p+2)}.$

(3) $L[1 - e^{at}] = \frac{1}{p} - \frac{1}{p-a} = \frac{-a}{p(p-a)}.$

(4) $L\left[\frac{ae^{at} - be^{bt}}{a-b}\right] = \frac{1}{a-b}\left(\frac{a}{p-a} - \frac{b}{p-b}\right) = \frac{p}{(p-a)(p-b)}.$

(5) $L\left[\frac{1}{b^2 - a^2}(\cos at - \sin bt)\right] = \frac{1}{b^2 - a^2}\left(\frac{p}{p^2 + a^2} - \frac{b}{p^2 + b^2}\right).$

(6) $L\left[\frac{at - \sin at}{a^3}\right] = \frac{1}{a^3}\left(\frac{a}{p^2} - \frac{a}{p^2 + a^2}\right) = \frac{1}{p^2(p^2 + a^2)}.$

(7) 由位移定理, $L[e^{-2t}\sin 5t] = \frac{5}{(p+2)^2 + 5^2}.$

$L[e^{p_0 t} f(t)] = F(p - p_0)$
 $= L[f(t)]|_{p=p-p_0}.$

(8) $L[e^{-(3+4i)t}] = \frac{1}{p + (3 + 4i)}.$

(9) 由位移定理, $L[te^{5t}] = \frac{1}{(p-5)^2}.$

(10) $L[\cosh \omega t] = L[\cos(i\omega t)] = \frac{p}{p^2 + (i\omega)^2} = \frac{p}{p^2 - \omega^2}.$

(11) 由位移定理, $L[e^{-at}\cos(\omega t + \varphi)] = \frac{(p+a)\cos \varphi - \omega \sin \varphi}{(p+a)^2 + \omega^2}.$

(12) 由本函数的微分法及位移定理,

$$\begin{aligned} L\left[\frac{d^2}{dt^2}(e^{-at}\sin \omega t)\right] &= p^2 L[e^{-at}\sin \omega t] - p \cdot 0 - \lim_{t \rightarrow 0^+} \frac{d}{dt}(e^{-at}\sin \omega t) \\ &= \frac{\omega p^2}{(p+a)^2 + \omega^2} - \omega. \end{aligned}$$

(13) 由位移定理, $L[t^2 e^t] = \frac{2!}{(p-1)^3} = \frac{2}{(p-1)^3}.$

(14) 由本函数的积分法及位移定理,

$$L\left[\int_0^\infty te^{2t} dt\right] = \frac{1}{p} L[te^{2t}] = \frac{1}{p(p-2)^2}.$$

(15) 记 $f(t) = \sinh 3t, g(t) = \sin 2t$, 由卷积定理,

$$L[f * g] = L[f] \cdot L[g] = \frac{3}{p^2 - 9} \cdot \frac{2}{p^2 + 4} = \frac{6}{(p^2 - 9)(p^2 + 4)}.$$

(16) 记 $f(t) = t^n, g(t) = e^{-at}\cos \omega t$, 由卷积定理及位移定理,

$$L[f_1 * f_2] = L[f_1] \cdot L[f_2] = \frac{n!}{p^{n+1}} \cdot \frac{p+a}{(p+a)^2 + \omega^2} = \frac{n!(p+a)}{p^{n+1}[(p+a)^2 + \omega^2]}.$$

(17) 由延迟定理, $L[\cos \omega(t - \varphi)h(t - \varphi)] = e^{-p\varphi} L[\cos \omega t] = \frac{pe^{-p\varphi}}{p^2 + \omega^2}.$

(18) 由延迟定理, $L[\cos \omega(t - \varphi)h(t - 2\varphi)] = e^{-2\varphi}L[\cos \omega(t + \varphi)] = e^{-2\varphi} \frac{p \cos \varphi - \omega \sin \varphi}{p^2 + \omega^2}$.

2. a

3. (1) $f(t) = (t - T)h(t - T)$, 因此像函数 $L[f(t)] = \frac{e^{-pT}}{p^2}$.

(2) $f(t) = -Eh(t - T)$, 因此像函数 $L[f(t)] = -\frac{Ee^{-pT}}{p}$.

(3) $f(t) = E[h(t - t_1) - h(t - t_1 - \tau_1)] + E[h(t - t_2) - h(t - t_2 - \tau_2)]$, 因此像函数

$$L[f(t)] = \frac{E}{p}e^{-pt_1}(1 - e^{-p\tau_1}) + \frac{E}{p}e^{-pt_2}(1 - e^{-p\tau_2}).$$

(4) $f(t) = Eh(t) - \frac{E}{4}[h(t - T) + h(t - 2T) + h(t - 3T) + h(t - 4T)]$, 因此像函数

$$L[f(t)] = \frac{E}{4p}(4 - e^{-pT} - e^{-2pT} - e^{-3pT} - e^{-4pT}).$$

(5) $f(t) = \frac{E}{T}t[h(t) - h(t - T)]$, 因此像函数 $L[f(t)] = \frac{E}{p^2T}[1 - (1 + pT)e^{-pT}]$.

4. a

5. 本题中四个函数均为周期函数.

(1) 周期为 T , 本函数 $f(t) = E\left(1 - \frac{2t}{T}\right)$ ($0 \leq t < T$), 因此其像函数为

$$\begin{aligned} L[f(t)] &= \frac{1}{1 - e^{-pT}} \int_0^T E\left(1 - \frac{2t}{T}\right)e^{-pt} dt = \frac{E}{1 - e^{-pT}} \cdot \frac{2 - p(T - 2t)}{p^2T} e^{-pt} \Big|_0^T \\ &= \frac{E}{p} \frac{1 + e^{-pT}}{1 - e^{-pT}} - \frac{2E}{p^2T} = \frac{E}{p} \tanh\left(\frac{1}{2}pT\right) - \frac{2E}{p^2T}. \end{aligned}$$

(2) 周期为 $\frac{2\pi}{\omega}$, 本函数 $f(t) = \sin \omega t[h(t) - h(t - \pi/\omega)]$ ($0 \leq t < 2\pi/\omega$), 因此其像函数为

$$\begin{aligned} L[f(t)] &= \frac{1}{1 - e^{-2p\pi/\omega}} \int_0^{\pi/\omega} A \sin \omega t e^{-pt} dt \\ &= \frac{A}{1 - e^{-2p\pi/\omega}} \cdot \frac{-1}{p^2 + \omega^2} e^{-pt} (\omega \cos \omega t + p \sin \omega t) \Big|_0^{\pi/\omega} \\ &= \frac{A\omega}{p^2 + \omega^2} \frac{1 + e^{-p\pi/\omega}}{1 - e^{-2p\pi/\omega}} = \frac{A\omega}{(p^2 + \omega^2)(1 - e^{-p\pi/\omega})}. \end{aligned}$$

(3) 周期为 T , 本函数 $f(t) = \begin{cases} E(1 - 2t/5T), & 0 \leq t < T/2, \\ 2E/5(t/T - 1), & T/2 \leq t < T, \end{cases}$ 因此其像函数为

$$\begin{aligned} L[f(t)] &= \frac{1}{1 - e^{-pT}} \left[\int_0^{T/2} E \left(1 - \frac{2t}{5T} \right) e^{-pt} dt + \int_{T/2}^T \frac{2E}{5} \left(\frac{t}{T} - 1 \right) e^{-pt} dt \right] \\ &= \frac{E}{1 - e^{-pT}} \left[\frac{(-2 + 5pT) + 2(1 - 2pT)e^{-pT/2}}{5p^2T} + \frac{(2 - pT)e^{-pT/2} - 2e^{-pT}}{5p^2T} \right] \\ &= \frac{E}{1 - e^{-pT}} \left[\frac{1}{p}(1 - e^{-pT/2}) - \frac{2}{5p^2T}(1 - e^{-pT/2})^2 \right] \\ &= \frac{E}{p} \frac{1 - e^{-pT/2}}{1 - e^{-pT}} \left[1 - \frac{2(1 - e^{-pT/2})}{5pT} \right]. \end{aligned}$$

(4) 周期为 T , 本函数 $f(t) = E \sin \frac{2\pi}{\tau} t [h(t) - h(t - \tau)]$ ($0 \leq t < T$), 因此其像函数为

$$\begin{aligned} L[f(t)] &= \frac{1}{1 - e^{-pT}} \int_0^\tau \sin \frac{2\pi t}{\tau} e^{-pt} dt \\ &= \frac{\omega=2\pi/\tau}{1 - e^{-pT}} \frac{E}{p^2 + \omega^2} \frac{-1}{\omega} e^{-pt} (\omega \cos \omega t + p \sin \omega t) \Big|_0^\tau \\ &= \frac{\omega E}{p^2 + \omega^2} \frac{1 - e^{-p\tau}}{1 - e^{-pT}} = \frac{2\pi\tau E}{p^2\tau^2 + 4\pi^2} \frac{1 - e^{-p\tau}}{1 - e^{-pT}}. \end{aligned}$$

6. 由拉式变换的线性性及位移定理,

$$L[f(t) \sin \omega t] = \frac{1}{2i} L[e^{i\omega t} f(t)] - \frac{1}{2i} L[e^{-i\omega t} f(t)] = \frac{1}{2i} [F(p - i\omega) - F(p + i\omega)].$$

$$7. (1) L^{-1} \left[\frac{1}{(p+3)(p+1)} \right] = \frac{1}{2} L^{-1} \left[\frac{1}{p+1} \right] - \frac{1}{2} L^{-1} \left[\frac{1}{p+3} \right] = \frac{1}{2} (e^{-t} - e^{-3t}).$$

$$(2) L^{-1} \left[\frac{1-p}{p^3+p^2+p+1} \right] = L^{-1} \left[\frac{1}{p+1} \right] - L^{-1} \left[\frac{p}{p^2+1} \right] = e^{-t} - \cos t.$$

$$(3) L^{-1} \left[\frac{p+2}{p^2+4p+5} \right] = L^{-1} \left[\frac{p+2}{(p+2)^2+1} \right] = e^{-2t} \cos t.$$

$$(4) L^{-1} \left[\frac{1}{p(p+a)} \right] = \frac{1}{a} L^{-1} \left[\frac{1}{p} \right] - \frac{1}{a} L^{-1} \left[\frac{1}{p+a} \right] = \frac{1 - e^{-at}}{a}.$$

$$(5) L^{-1} \left[\frac{1}{p(p-1)(p-2)} \right] = \frac{1}{2} L^{-1} \left[\frac{1}{p-2} \right] - L^{-1} \left[\frac{1}{p-1} \right] + \frac{1}{2} L^{-1} \left[\frac{1}{p} \right] = \frac{1}{2} (e^t - 1)^2.$$

$$(6) L^{-1} \left[\frac{1}{(p^2+1)(p^2+3)} \right] = \frac{1}{2} L^{-1} \left[\frac{1}{p^2+1} \right] - \frac{1}{2} L^{-1} \left[\frac{1}{p^2+3} \right] = \frac{\sqrt{3} \sin t - \sin \sqrt{3}t}{2\sqrt{3}}.$$

(7) 先将 $\frac{1}{(p-2)p}$ 拆分, 再分别将 $\frac{1}{(p-2)(p^2+1)}$ 和 $\frac{1}{p(p^2+1)}$ 拆分, 得

$$\begin{aligned} L^{-1} \left[\frac{1}{p(p-2)(p^2+1)} \right] &= \frac{1}{10} L^{-1} \left[\frac{1}{p-2} \right] - \frac{1}{2} L^{-1} \left[\frac{1}{p} \right] + \frac{1}{5} L^{-1} \left[\frac{2p-1}{p^2+1} \right] \\ &= \frac{1}{10} e^{2t} + \frac{1}{5} (2 \cos t - \sin t) - \frac{1}{2}. \end{aligned}$$

(8) 由位移定理,

$$\begin{aligned} L^{-1}\left[\frac{1}{p(p-2)^2}\right] &= \frac{1}{4}L^{-1}\left[\frac{1}{p}\right] - \frac{1}{4}L^{-1}\left[\frac{1}{p-2}\right] + \frac{1}{2}L^{-1}\left[\frac{1}{(p-2)^2}\right] \\ &= \frac{1}{4}[(2t-1)e^{2t} + 1]. \end{aligned}$$

(9) 由位移定理,

$$\begin{aligned} L^{-1}\left[\frac{p+3}{p^3+3p^2+6p+4}\right] &= L^{-1}\left[\frac{1}{p+1}\right] - L^{-1}\left[\frac{(p+1)-1}{(p+1)^2+3}\right] \\ &= e^{-t} - e^{-t}\left(\cos\sqrt{3}t - \frac{1}{\sqrt{3}}\sin\sqrt{3}t\right) \\ &= e^{-t}\left[1 - \frac{2}{\sqrt{3}}\cos\left(\sqrt{3}t + \frac{\pi}{6}\right)\right]. \end{aligned}$$

$$(10) \quad L^{-1}\left[\frac{p}{p^4+3p^2-4}\right] = \frac{1}{5}L^{-1}\left[\frac{p}{p^2-1}\right] - \frac{1}{5}L^{-1}\left[\frac{p}{p^2+4}\right] = \frac{1}{5}(\cosh t - \cos 2t).$$

(11) 由卷积定理,

$$\begin{aligned} L^{-1}\left[\frac{1}{p^4-3p^3+3p^2-p}\right] &= L^{-1}\left[\frac{1}{p}\right] * L^{-1}\left[\frac{1}{(p-1)^3}\right] = \int_0^t \frac{1}{2}x^2e^x dx \\ &= \frac{1}{2}(t^2-2t+2)e^t - 1. \end{aligned}$$

(12) 将分母拆解成二次项:

$$\begin{aligned} L^{-1}\left[\frac{a^2p}{p^4+a^4}\right] &= \frac{1}{2i}L^{-1}\left[\frac{p}{p^2-ia^2}\right] - \frac{1}{2i}L^{-1}\left[\frac{p}{p^2+ia^2}\right] \\ &= \frac{1}{2i}\cos(e^{-\pi i/4}at) - \frac{1}{2i}\cos(e^{\pi i/4}at) \\ &= \frac{1}{i}\sin\left(\frac{e^{\pi i/4}+e^{-\pi i/4}}{2}at\right)\sin\left(\frac{e^{\pi i/4}-e^{-\pi i/4}}{2}at\right) \\ &= \sin\left(\frac{at}{\sqrt{2}}\right)\sinh\left(\frac{at}{\sqrt{2}}\right). \end{aligned}$$

(13) 仿照上一问的解法,

$$\begin{aligned} L^{-1}\left[\frac{p^3}{p^4+a^4}\right] &= \frac{1}{2i}\cos(e^{-\pi i/4}at) - \frac{1}{2i}\cos(e^{\pi i/4}at) \\ &= \frac{1}{i}\cos\left(\frac{e^{\pi i/4}+e^{-\pi i/4}}{2}at\right)\cos\left(\frac{e^{\pi i/4}-e^{-\pi i/4}}{2}at\right) \\ &= \cos\left(\frac{at}{\sqrt{2}}\right)\cosh\left(\frac{at}{\sqrt{2}}\right). \end{aligned}$$

$$(14) \quad \text{由位移定理, } L^{-1}\left[\frac{1}{(p+1)^4}\right] = \frac{1}{6}t^3e^{-t}.$$

(15) 由像函数的微分法及位移定理,

$$\begin{aligned} L^{-1}\left[\frac{p-1}{(p^2-2p+2)^2}\right] &= L^{-1}\left[-\frac{1}{2}\frac{d}{dp}\left(\frac{1}{(p-1)^2+1}\right)\right] \\ &= \frac{t}{2}L^{-1}\left[\frac{1}{(p-1)^2+1}\right] = \frac{t}{2}e^t\sin t. \end{aligned}$$

$$(16) \quad L^{-1} \left[\frac{3p+7}{p^2+2p+1+a^2} \right] = L^{-1} \left[\frac{3(p+1)+4}{(p+1)^2+a^2} \right] = e^{-t} \left(3 \cos at + \frac{4}{a} \sin at \right).$$

(17) 由延迟定理, e^{-p} 对应将 $\frac{p+2}{p^2+1}$ 的本函数中 t 换成 $(t-1)$, 因此

$$L^{-1} \left[\frac{p+2}{p^2+1} e^{-p} \right] = L^{-1} \left[\frac{p+2}{p^2+1} \right] \Big|_{t \rightarrow t-1} = [\cos(t-1) + 2 \sin(t-1)] h(t-1).$$

(18) 由延迟定理, e^{-10p} 将 $\frac{1-p}{(p+1)(p^2+1)}$ 的本函数中 t 换成 $(t-10)$, 因此

$$\begin{aligned} L^{-1} \left[\frac{1-p}{(p+1)(p^2+1)} e^{-10p} \right] &= L^{-1} \left[\frac{1}{p+1} \right] - L^{-1} \left[\frac{p}{p^2+1} \right] \Big|_{t \rightarrow t-10} \\ &= [e^{-(t-10)} - \cos(t-10)] h(t-10). \end{aligned}$$

(19) 由延迟定理, $L^{-1} \left[\frac{1-e^{-3p}}{p} \right] = h(t) - L^{-1} \left[\frac{1}{p} \right] \Big|_{t \rightarrow t-3} = h(t) - h(t-3).$

(20) 将 $\frac{1}{1+e^{-p\pi}}$ 在 $\operatorname{Re}(p) = \infty$ 附近展开, 得 $\frac{1}{1+e^{-p\pi}} = \sum_{n=0}^{\infty} e^{-pn\pi}$, 由拉氏变换的线性性与延迟定理,

$$\begin{aligned} L^{-1} \left[\frac{p}{p^2+1} \right] &= \sum_{n=0}^{\infty} L^{-1} \left[e^{-pn\pi} \frac{p}{p^2+1} \right] = \sum_{n=0}^{\infty} \cos(t-n\pi) h(t-n\pi) \\ &= \cos t \sum_{n=0}^{\infty} (-1)^n h(t-n\pi) \\ 1 &= \begin{cases} \cos t, & 2n\pi \leq t < (2n+1)\pi, \\ 0, & (2n+1)\pi \leq t < 2(n+1)\pi \end{cases} \quad (n \in \mathbb{N}). \end{aligned}$$

8. 以下均记 $X(p) = L[x(t)], Y(p) = L[y(t)], Z(p) = L[z(t)].$

(1) $L[y''(t)] = p^2 Y(p), L[y'(t)] = pY(p)$, 那么原方程变为

$$p^2 Y(p) + pY(p) = L[1] = \frac{1}{t},$$

因此 $Y(p) = \frac{1}{p^2(p+1)}$, 故

$$y(t) = L^{-1}[Y(p)] = L^{-1} \left[\frac{1}{p^2} \right] - L^{-1} \left[\frac{1}{p} \right] + L^{-1} \left[\frac{1}{p+1} \right] = e^{-t} + t - 1.$$

(2) $L[y''(t)] = p^2 Y(p), L[y'(t)] = pY(p)$, 那么原方程变为

$$p^2 Y(p) - pY(p) = L[e^t] = \frac{1}{p-1},$$

因此 $Y(p) = \frac{1}{p(p-1)^2}$, 故

$$y(t) = L^{-1}[Y(p)] = L^{-1} \left[\frac{1}{(p-1)^2} \right] - L^{-1} \left[\frac{1}{p-1} \right] + L^{-1} \left[\frac{1}{p} \right] = (t-1)e^t + 1.$$

(3) $L[y''(0)] = p^2Y(p) - 1, L[y'(t)] = pY(p)$, 那么原方程变为

$$p^2Y(p) - (a+b)pY(p) + abY(p) = 1,$$

因此 $Y(p) = \frac{1}{(p-a)(p-b)}$, 故

$$\begin{aligned} y(t) &= L^{-1}[Y(p)] = \frac{1}{b-a}L^{-1}\left[\frac{1}{p-b}\right] - \frac{1}{b-a}L^{-1}\left[\frac{1}{p-a}\right] \\ &= \frac{e^{bt} - e^{at}}{b-a}. \end{aligned}$$

(4) $L[y''(t)] = p^2Y(p), L[y'(t)] = pY(p)$, 那么原方程变为

$$p^2Y(p) - 2pY(p) + Y(p) = L[te^t] = \frac{1}{(p-1)^2},$$

因此 $Y(p) = \frac{1}{(p-1)^4}$, 因此

$$y(t) = L^{-1}[Y(p)] = L^{-1}\left[\frac{1}{(p-1)^4}\right] = \frac{1}{6}t^3e^t.$$

(5) $L[y''(t)] = p^2Y(p) + p + 2$, 那么原方程变为

$$p^2Y(p) - Y(p) + p + 2 = \frac{4}{p^2+1} + \frac{5p}{p^2+4},$$

因此 $Y(p) = \frac{4}{(p^2-1)(p^2+1)} + \frac{5p}{(p^2-1)(p^2+4)} - \frac{p+2}{p^2-1}$, 故

$$y(t) = L^{-1}[Y(p)] = -2L^{-1}\left[\frac{1}{p^2+1}\right] - L^{-1}\left[\frac{p}{p^2+4}\right] = -2\sin t - \cos 2t.$$

(6) $L[y''(t)] = p^2Y(p)$, 那么原方程变为

$$p^2Y(p) - Y(p) = L[th(t) - (t-1)h(t-1)] = \frac{1-e^{-1}}{p^2},$$

因此 $Y(p) = \frac{1-e^{-1}}{p^2(p^2-1)}$, 故

$$\begin{aligned} y(t) &= L^{-1}[Y(p)] = L^{-1}\left[\frac{1-e^{-1}}{p^2-1}\right] - L^{-1}\left[\frac{1-e^{-1}}{p^2}\right] \\ &= \sinh th(t) - \sinh(t-1)h(t-1) - th(t) + (t-1)h(t-1) \\ &= (\sinh t - t)h(t) - [\sinh(t-1) - (t-1)]h(t-1). \end{aligned}$$

(7) $L[y^{(n)}(t)] = p^nY(p)$ ($n = 0, 1, 2, 3$), 那么原方程变为

$$p^3Y(p) + 3p^2Y(p) + 3pY(p) + Y(p) = L[6e^{-t}] = \frac{6}{p+1},$$

因此 $Y(p) = \frac{6}{(p+1)^4}$, 因此

$$y(t) = L^{-1}[Y(p)] = 6L^{-1}\left[\frac{1}{(p+1)^4}\right] = t^3e^{-t}.$$

(8) $L[x'(t)] = pX(p) - b, L[y'(t)] = pY(p) - a$, 那么原方程组变为

$$\begin{cases} pY(p) - a + pX(p) - b = 4Y(p) + 1/p, \\ pY(p) - a + X(p) = 3Y(p) + 2/p^3, \end{cases}$$

因此 $Y(p) = \frac{ap}{(p-2)^2} + \frac{2}{p^2(p-2)^2} - \frac{a+b}{(p-2)^2} - \frac{1}{p(p-2)^2}$, 故

$$\begin{aligned} y(t) &= L^{-1}[Y(p)] = L^{-1}\left[\frac{4a-1}{4(p-2)}\right] + L^{-1}\left[\frac{a-b}{(p-2)^2}\right] + L^{-1}\left[\frac{1}{4p}\right] + L^{-1}\left[\frac{1}{2p^2}\right] \\ &= \left(a - \frac{1}{4}\right)e^{2t} + (a-b)te^{2t} + \frac{1}{2}t + \frac{1}{4} \\ x(t) &= 3y(t) - y'(t) + t^2 = \left(b - \frac{1}{4}\right)e^{2t} + (a-b)te^{2t} + t^2 + \frac{3}{2}t + \frac{1}{4}. \end{aligned}$$

(9) $L[x'(t)] = pX(p), L[y'(t)] = pY(p) - 1$, 那么原方程组变为

$$\begin{cases} pX(p) - 2pY(p) + 2 = 1/(p^2 + 1), \\ pX(p) + pY(p) - 1 = p/(p^2 + 1), \end{cases}$$

因此 $X(p) = \frac{1}{3}\left(\frac{1}{p} - \frac{p-2}{p^2+1}\right), Y(p) = \frac{1}{3}\left(\frac{2}{p} + \frac{p+1}{p^2+1}\right)$, 故

$$\begin{aligned} x(t) &= L^{-1}[X(p)] = \frac{1}{3}(1 - \cos t + 2 \sin t), \\ y(t) &= L^{-1}[Y(p)] = \frac{1}{3}(2 + \cos t + \sin t). \end{aligned}$$

(10) $L[x'(t)] = pX(p), L[y'(t)] = pY(p), L[z'(t)] = pZ(p)$, 那么原方程组变为

$$\begin{cases} pX(p) - pY(p) = 0, \\ pY(p) + pZ(p) = 1/p, \\ pX(p) - pZ(p) = 1/p^2, \end{cases}$$

因此 $X(p) = Y(p) = \frac{1}{2}\left(\frac{1}{p^2} + \frac{1}{p^3}\right), Z(p) = \frac{1}{2}\left(\frac{1}{p^2} - \frac{1}{p^3}\right)$, 故

$$x(t) = y(t) = \frac{1}{2}t + \frac{1}{4}t^2, \quad z(t) = \frac{1}{2}t - \frac{1}{4}t^2.$$

9. 记 $L[y(t)] = Y(p), L[f(t)] = F(p)$, 且 $y'(0) = y''(0) = 0$, 那么原方程变为

$$p^2Y(p) + \omega^2Y(p) = F(p),$$

因此 $Y(p) = \frac{F(p)}{p^2 + \omega^2}$, 结合卷积定理,

$$\begin{aligned} y(t) &= L^{-1}[Y(p)] = L^{-1}\left[\frac{1}{p^2 + \omega^2} \cdot F(p)\right] = L^{-1}\left\{L\left[\frac{1}{\omega} \sin \omega t\right] \cdot L[f(t)]\right\} \\ &= \frac{1}{\omega} \int_0^t \sin \omega(t-u) f(u) du. \end{aligned}$$

10. 记 $L[f(t)] = F(p)$, 由卷积定理, 原方程变为

$$F(p) = \frac{ab}{p^2 + b^2} + \frac{bc}{p^2 + b^2} F(p),$$

因此 $F(p) = \frac{ab}{p^2 + (b^2 - bc)}$, 故

$$f(t) = L^{-1}[F(p)] = \frac{ab}{\sqrt{b^2 - bc}} \sin \left(\sqrt{b^2 - bc} t \right).$$